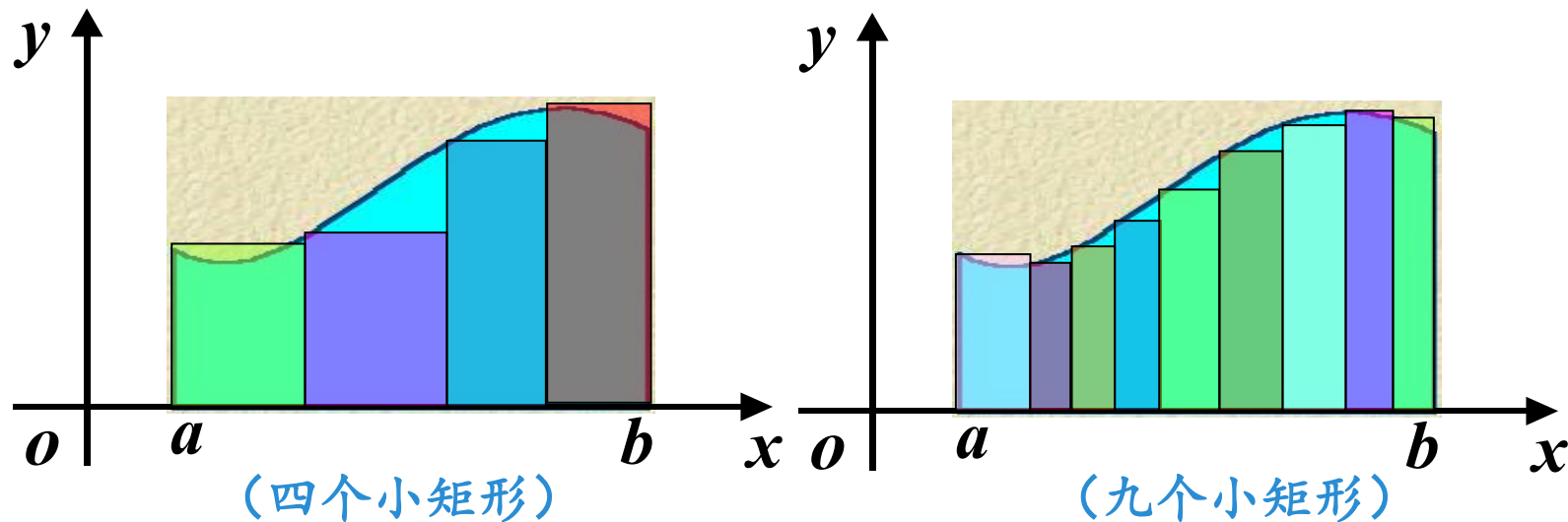


Chp24 定积分及其应用

Chp24.1 定积分的意义 Pg. 114

虽然曲边梯形的准确面积我们不会计算，但是我们可以用一些小矩形来近似算出它的面积。



从中可以得到一个什么样的启示？

小曲边梯形的底:

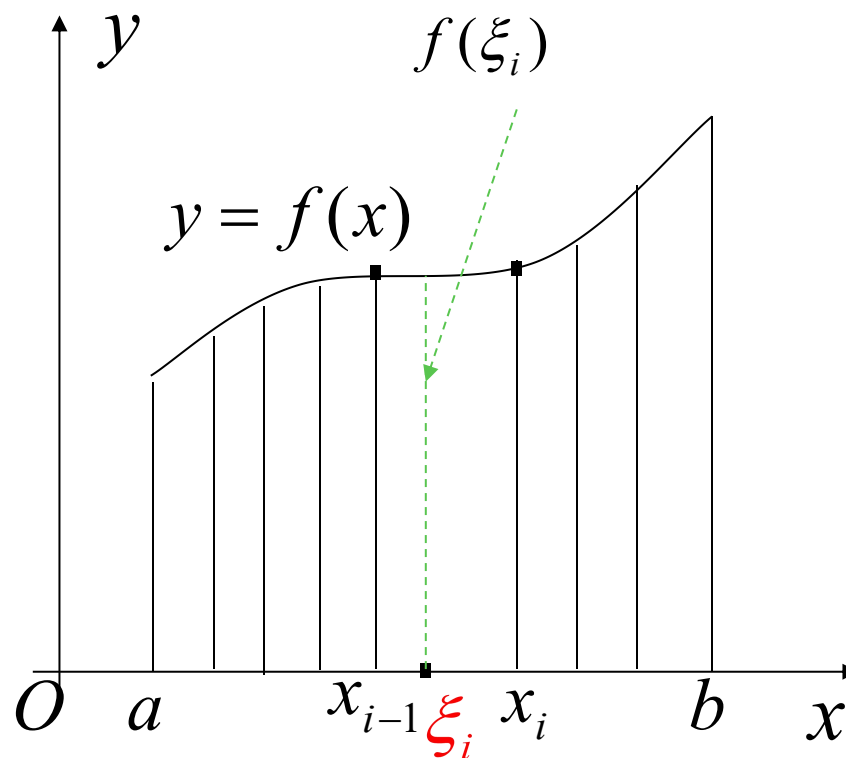
$$[x_{i-1}, x_i]$$

小曲边梯形的高:

$$f(\xi_i)$$

小曲边梯形的面积:

$$\Delta A_i \approx f(\xi_i)(x_i - x_{i-1})$$



(1) 分割

用任意的一组分点：

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$$

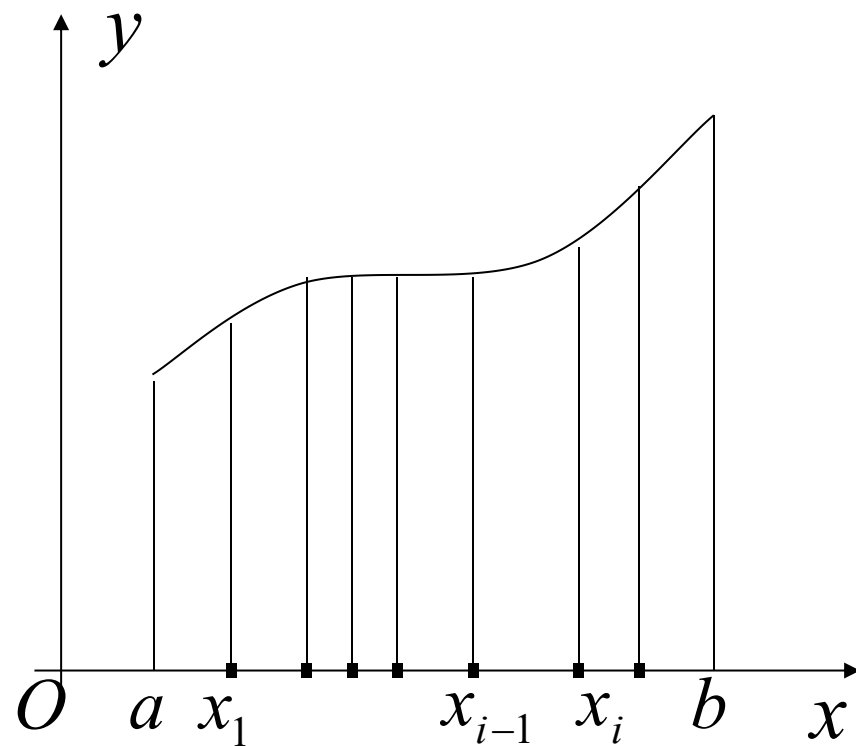
把 $[a, b]$ 分成 n 个小区间

$$[x_{i-1}, x_i] \quad i=1, 2, \dots, n$$

相应地把曲边梯形分为 n

个小曲边梯形，其面积分

别记为 $\Delta A_i \quad i=1, 2, \dots, n$



(2) 近似代替

在每个小区间 $[x_{i-1}, x_i]$ 上

任取一点 ξ_i ,

于是小曲边梯形的面积

$$\Delta A_i \approx f(\xi_i) \Delta x_i$$

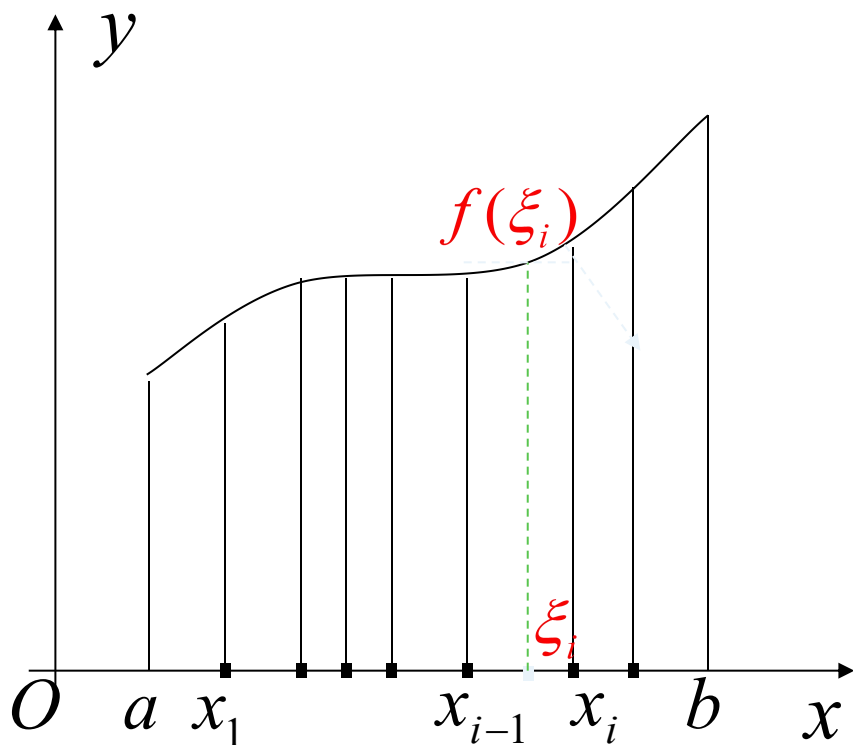
其中

$$\Delta x_i = x_i - x_{i-1}$$

(3) 求和

大曲边梯形的面积

$$A \approx \sum_{i=1}^n \Delta A_i \approx \sum_{i=1}^n f(\xi_i) \Delta x_i$$



(4) 取极限

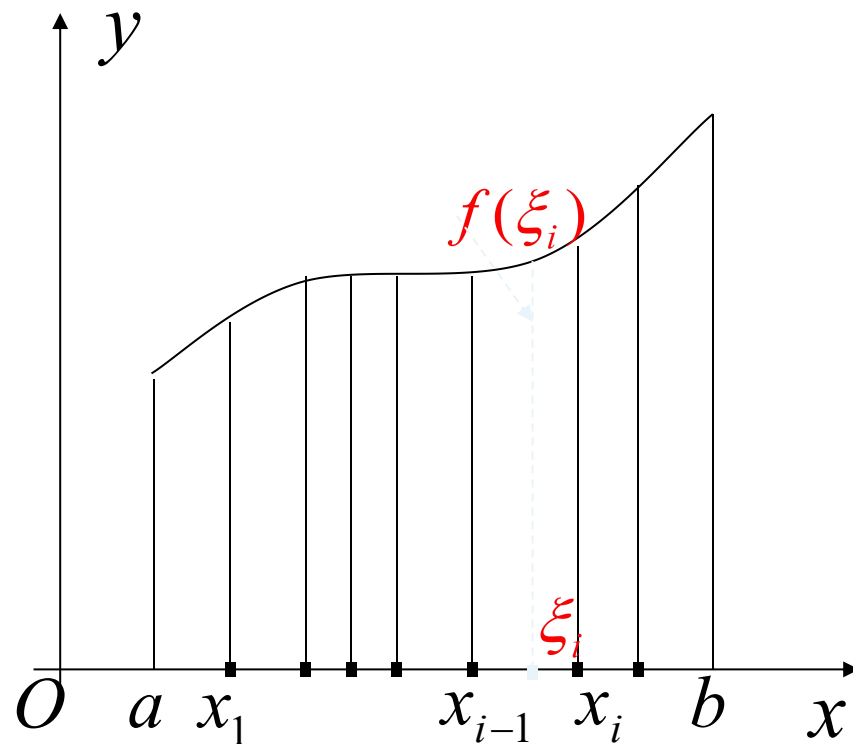
让每个小区间的长度趋近于零,

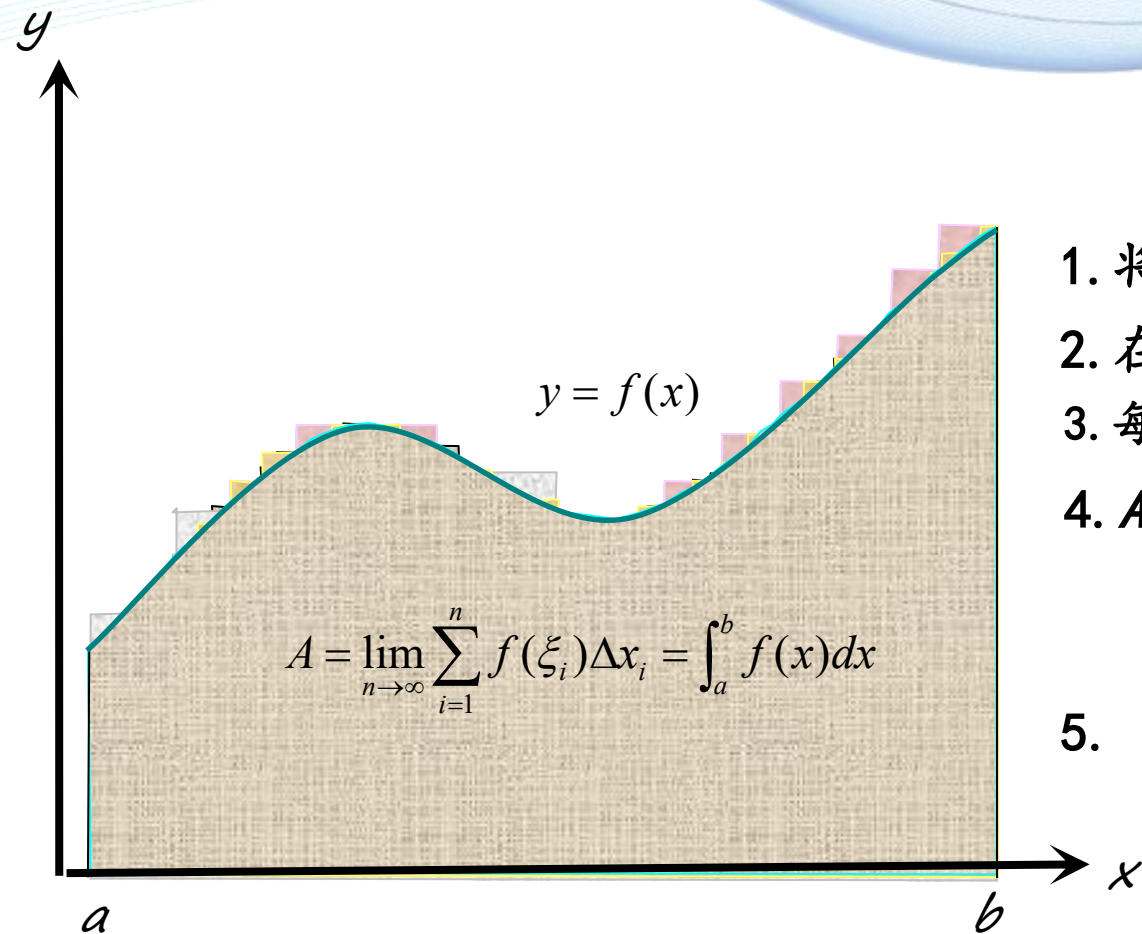
即小曲边梯形趋近于无穷多

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta A_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x_i$$

则定义此极限值为曲边梯形的面积

记作 $A = \int_a^b f(x) dx$





1. 将 $[a, b]$ 分割为 n 个小区间
2. 在每个小区间上任取一点 x_i
3. 每个小区间上的曲线 $y=f(x)$ 用直线段 $y=f(x_i)$ 代替
4. $A_{\Delta} = f(\xi_1)\Delta x_1 + f(\xi_2)\Delta x_2 + \cdots + f(\xi_i)\Delta x_i + \cdots + f(\xi_n)\Delta x_n$
 $= \sum_{i=1}^n f(\xi_i)\Delta x_i \quad (\Delta x_i = x_i - x_{i-1})$
5. $A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta A_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i)\Delta x_i$



$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x$$

$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x$ 叫做函数 $f(x)$ 由 a 至 b 的定积分, 以 $\int_a^b f(x) dx$ 表示

$\sum_{i=1}^n f(\xi_i) \Delta x$: 黎曼和 (*Riemann sum*), 用以纪念德国数学家黎曼

$f(x)$: 被积函数 (*integrand*)

$\int$: 积分符号 (*integral sign*)

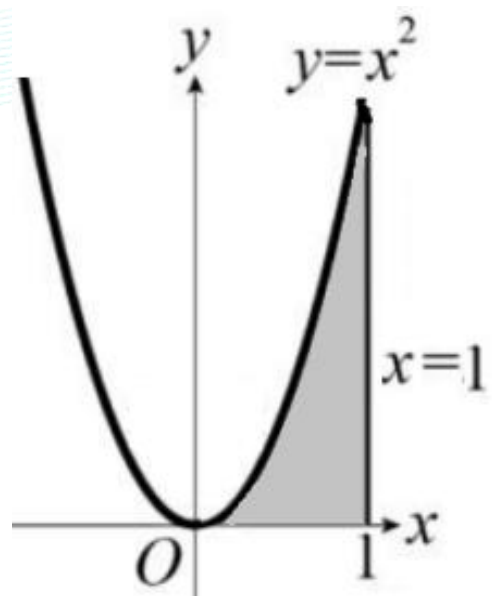
dx : 积分变量 (*variable of integration*)

$[a, b]$: 积分区间

a : 积分下限 (*lower limit*)

b : 积分上限 (*upper limit*)

a) 利用定积分定义, 求抛物线 $y = x^2$, 直线 $x = 1$ 以及 x 轴所围成的曲边三角形面积(如图)。



将区间 $[0, 1]$ 平分成 n 等分即 $\left[0, \frac{1}{n}\right], \left[\frac{1}{n}, \frac{2}{n}\right], \dots, \left[\frac{i-1}{n}, \frac{i}{n}\right], \dots, \left[\frac{n-1}{n}, 1\right]$,

其中宽度为 $\Delta x = \frac{1}{n}$

思考:

若将区间 $[1, 2]$ 平分 n 等分,

其 n 个区间为 _____

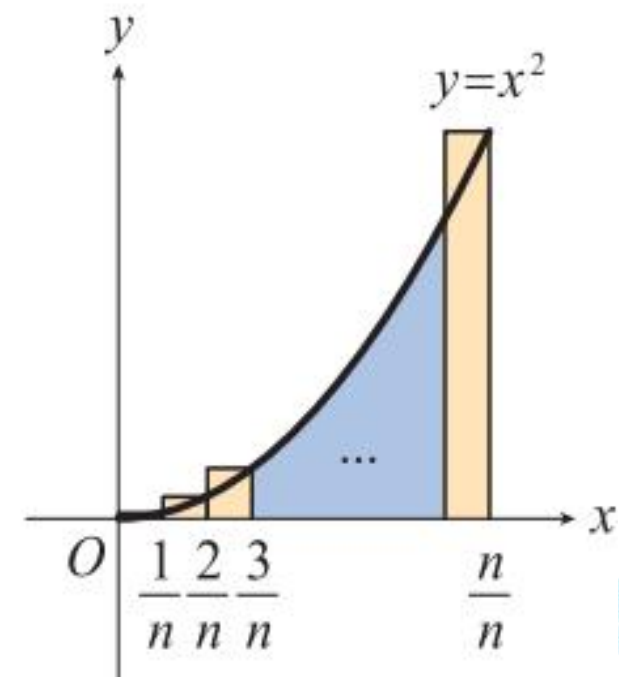
每个小区间 $\left[\frac{i-1}{n}, \frac{i}{n}\right]$ 取右端点为 ξ_i 作为高, 即 ξ_i 为 $\frac{i}{n}$, 其中 $i = 1, 2, \dots, n$

每个小矩形面积 $\Delta A_i = f(\xi_i) \Delta x$

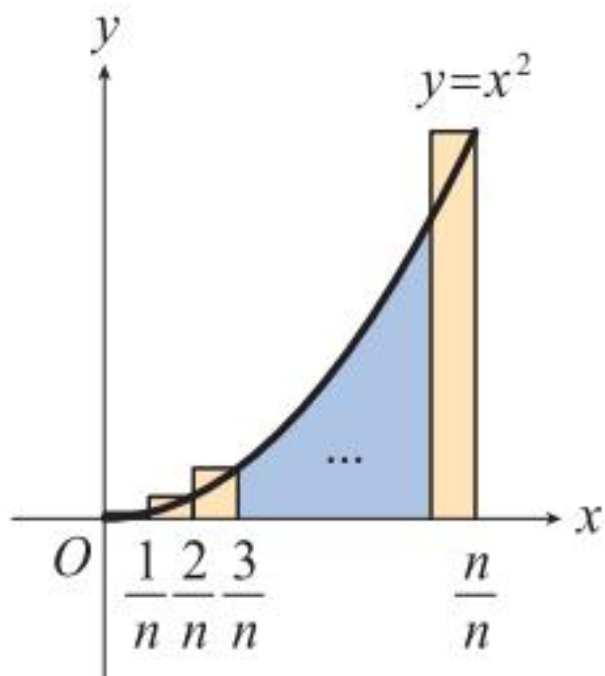
$$= \xi_i^2 \Delta x$$

$$= \left(\frac{i}{n}\right)^2 \frac{1}{n}$$

$$= \frac{i^2}{n^3}$$



a) 利用定积分定义, 求抛物线 $y = x^2$, 直线 $x = 1$ 以及 x 轴所围成的曲边三角形面积(如图)。



$$A \approx \Delta A_1 + \dots + \Delta A_n$$

$$\approx \sum_{i=1}^n \Delta A_i$$

$$\approx \sum_{i=1}^n \frac{i^2}{n^3}$$

$$\approx \frac{1^2}{n^3} + \frac{2^2}{n^3} + \dots + \frac{n^2}{n^3}$$

$$\approx \frac{1}{n^3} \left(\frac{n(n+1)(2n+1)}{6} \right)$$

$$\approx \frac{2n^2 + 3n + 1}{6n^2}$$

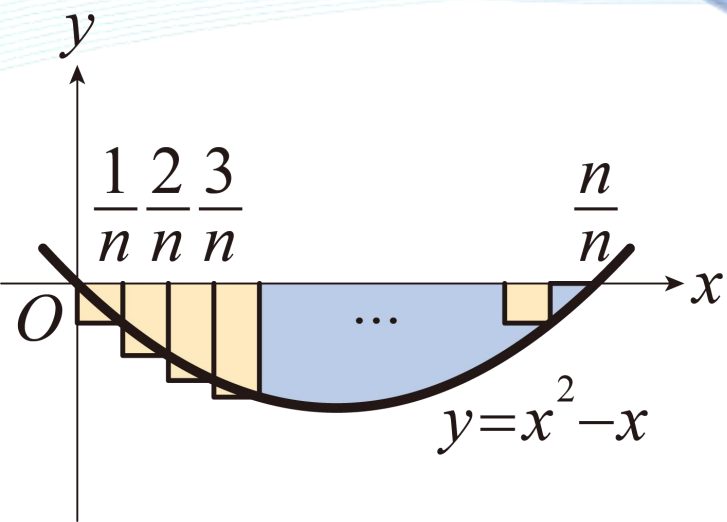
$$\begin{aligned} A &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta A_i \\ &= \lim_{n \rightarrow \infty} \frac{2n^2 + 3n + 1}{6n^2} \\ &= \lim_{n \rightarrow \infty} \frac{2 + \frac{3}{n} + \frac{1}{n^2}}{6} \\ &= \frac{1}{3} (\text{平方单位}) \end{aligned}$$

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x$$

若题目没有限制用积分定义, 则用

$$\int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3} - 0 = \frac{1}{3}$$

b) 利用定积分定义, 求 $\int_0^1 (x^2 - x) dx$ 。



将区间 $[0, 1]$ 平分成 n 等分即 $\left[0, \frac{1}{n}\right], \left[\frac{1}{n}, \frac{2}{n}\right], \dots, \left[\frac{i-1}{n}, \frac{i}{n}\right], \dots, \left[\frac{n-1}{n}, 1\right]$,

其中宽度为 $\Delta x = \frac{1}{n}$

每个小区间 $\left[\frac{i-1}{n}, \frac{i}{n}\right]$ 取右端点为 ξ_i 作为高, 即 ξ_i 为 $\frac{i}{n}$, 其中 $i = 1, 2, \dots, n$

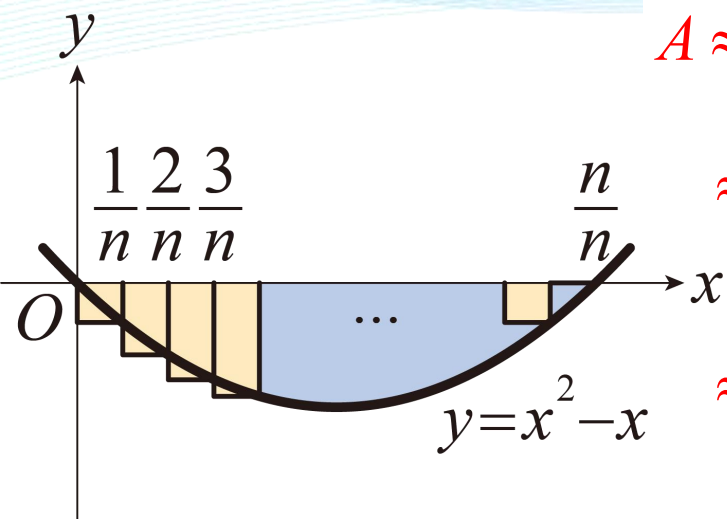
每个小矩形面积 $\Delta A_i = f(\xi_i) \Delta x$

$$= \left(\xi_i^2 - \xi_i \right) \Delta x$$

$$= \left[\left(\frac{i}{n} \right)^2 - \frac{i}{n} \right] \frac{1}{n}$$

$$= \frac{i^2}{n^3} - \frac{i}{n^2}$$

b) 利用定积分定义, 求 $\int_0^1 (x^2 - x) dx$.



$$A \approx \Delta A_1 + \dots + \Delta A_n$$

$$\approx \sum_{i=1}^n \Delta A_i$$

$$\approx \sum_{i=1}^n \left(\frac{i^2}{n^3} - \frac{i}{n^2} \right)$$

$$\approx \left(\frac{1^2}{n^3} + \frac{2^2}{n^3} + \dots + \frac{n^2}{n^3} \right) - \left(\frac{1}{n^2} + \frac{2}{n^2} + \dots + \frac{n}{n^2} \right)$$

$$\approx \frac{1}{n^3} \left(\frac{n(n+1)(2n+1)}{6} \right) - \frac{1}{n^2} \left(\frac{n(n+1)}{2} \right)$$

$$\approx \frac{2n^2 + 3n + 1}{6n^2} - \frac{n+1}{2n}$$

$$\approx \frac{-n^2 + 1}{6n^2}$$

$$\int_0^1 (x^2 - x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta A_i$$

$$= \lim_{n \rightarrow \infty} \frac{-n^2 + 1}{6n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{-1 + \frac{1}{n}}{6}$$

$$= -\frac{1}{6}$$

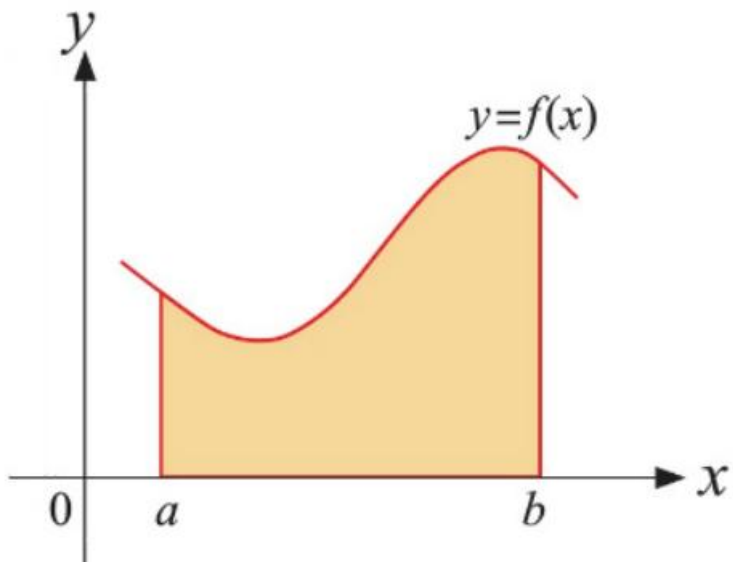
由上可见,
定积分的值为负值

若题目没有限制用积分定义,
则用

$$\int_0^1 (x^2 - x) dx = \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_0^1 = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6}$$

定积分

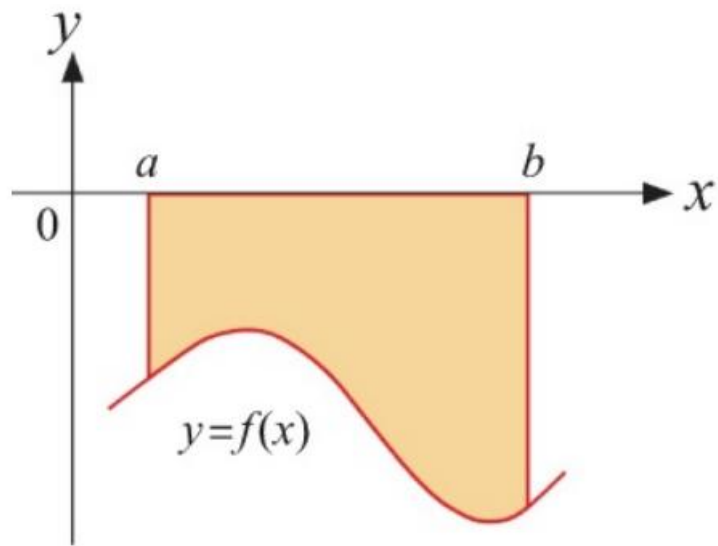
面积



$$\because f(x) > 0$$

$$\therefore \int_a^b f(x) dx > 0 (\text{正值})$$

$$A = \int_a^b f(x) dx$$



$$\because f(x) < 0$$

$$\therefore \int_a^b f(x) dx < 0 (\text{负值})$$

$$A = \left| \int_a^b f(x) dx \right| \text{ 或 } -\int_a^b f(x) dx$$



定积分 $\int_a^b f(x) dx$ 是一个定值，与自变量 x 无关。

所以我們也可將定積分 $\int_a^b f(x) dx$ 表示為 $\int_a^b f(u) du$ 或 $\int_a^b f(t) dt$ 。

但不定積分 $\int f(x) dx$ 是 x 的函數， $\int f(u) du$ 是 u 的函數，二者不同。

微积分基本定理(*Fundamental Theorem of Calculus*)的第一部分

若函数 f 在区间 $[a, b]$ 上连续, 则可定义函数 g

$$g(x) = \int_a^x f(t) dt, a \leq x \leq b$$

其中函数 g 在区间 $[a, b]$ 上连续且 $g'(x) = f(x)$

其相关例题放置章节最后部分才上

已知 $f(x)$ 为在区间 $[a, b]$ 上连续的函数, 令函数 $g(x) = \int_a^x f(t)dt$, 其中 $a \leq x \leq b$ 。

明显可知函数 g 随着定积分的上限 x 而变。若 x 为固定值, 则定积分 $\int_a^x f(t)dt$ 也为定值; 若 x 为变量, 则定积分 $\int_a^x f(t)dt$ 也随之而变, 由此可定义出一个以 x 为变数的函数 $g(x)$ 。

若在区间 $[a, b]$ 上 $f \geq 0$, 则 $g(x)$ 可视为上限为 x 、下限为 a 的曲边梯形的面积。以下我们以微分的定义计算函数 $g(x)$ 的导数。

如图 24-5 所示, 若 h 为一个很小的正数, 则二曲边梯形 $g(x+h)$ 及 $g(x)$ 的面积差

$$mh \leq g(x+h) - g(x) \leq Mh$$

其中 m 及 M 分别为 f 在区间 $[x, x+h]$ 上的最小值及最大值。因此,

$$m \leq \frac{g(x+h) - g(x)}{h} \leq M$$

显然, 当 h 趋近于 0, m 及 M 都趋近于 $f(x)$ 。因此,

$$\begin{aligned} \frac{d}{dx} g(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = f(x) \\ \therefore g'(x) &= f(x) \end{aligned}$$

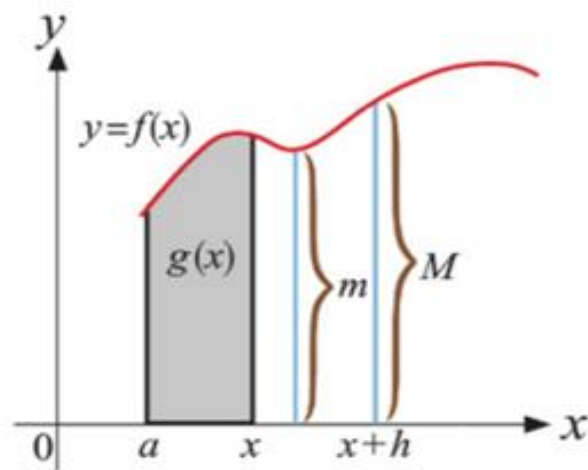


图24-5

微积分基本定理(*Fundamental Theorem of Calculus*)的第二部分

若函数 f 在区间 $[a, b]$ 上连续, 则有

$$\int_a^b f(x) dx = F(b) - F(a)$$

其中 F 为 f 的任意一个原函数, 即 $F'(x) = f(x)$

证：令 $g(x) = \int_a^x f(t)dt$ ，由 $g'(x) = f(x)$ 可知 $g(x)$ 为 $f(x)$ 的其中一个原函数。令 $F(x)$ 为 $f(x)$ 的另一个原函数，则有

$$F'(x) - g'(x) = f(x) - f(x) = 0$$

这表示 $\int [F'(x) - g'(x)]dx = c$ ，因此， $f(x)$ 的任意两个原函数相差一常数：

$$F(x) - g(x) = c, \quad a \leq x \leq b$$

由此可得，

$$\begin{aligned} F(b) - F(a) &= [g(b) + c] - [g(a) + c] \\ &= g(b) - g(a) \\ &= \int_a^b f(t)dt - \int_a^a f(t)dt \\ &= \int_a^b f(x)dx \end{aligned}$$

微积分基本定理第二部分表示，若要计算定积分 $\int_a^b f(x)dx$ 的值，只需要知道 $f(x)$ 任一原函数 $F(x)$ 在积分区间两端点的函数值 $F(a)$ 及 $F(b)$ 即可。一般上，我们将它写成以下的形式：

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$$

从上述证明可知，无论用哪个 $f(x)$ 的原函数来计算，定积分 $\int_a^b f(x)dx$ 的值都是一样的。

c)求下列定积分:

$$i) \int_0^1 x^2 dx$$

$$\begin{aligned} & \int_0^1 x^2 dx \\ &= \left[\frac{x^3}{3} + C \right]_0^1 \\ &= \left(\frac{1}{3} + C \right) - (0 + C) \\ &= \frac{1}{3} \end{aligned}$$

$$ii) \int_1^4 \frac{dx}{\sqrt{x}}$$

$$\begin{aligned} & \int_1^4 x^{-\frac{1}{2}} dx \\ &= \left[\frac{x^{\frac{1}{2}}}{\frac{1}{2}} \right]_1^4 \\ &= \left[2\sqrt{x} \right]_1^4 \\ &= (2\sqrt{4}) - (2\sqrt{1}) \\ &= 2 \end{aligned}$$

$$iii) \int_0^1 dx$$

$$\begin{aligned} &= \int_0^1 1 dx \\ &= [x]_0^1 \\ &= 1 - 0 \\ &= 1 \end{aligned}$$

d) 已知 $y = \left(\frac{x}{1+x} \right)^5$, 证明 $\frac{dy}{dx} = \frac{5x^4}{(1+x)^6}$ 。据此, 求 $\int_0^1 \frac{x^4}{(1+x)^6} dx$ 。

$$y = \left(\frac{x}{1+x} \right)^5$$

$$\begin{aligned} \frac{dy}{dx} &= 5 \left(\frac{x}{1+x} \right)^4 \left(\frac{(1+x) - x}{(1+x)^2} \right) \\ &= \frac{5x^4}{(1+x)^6} \quad \text{得证} \end{aligned}$$

$$\therefore \int \frac{5x^4}{(1+x)^6} dx = \left(\frac{x}{1+x} \right)^5 + C$$

$$\int_0^1 \frac{x^4}{(1+x)^6} dx$$

$$= \frac{1}{5} \int_0^1 \frac{5x^4}{(1+x)^6} dx$$

$$= \frac{1}{5} \left[\left(\frac{x}{1+x} \right)^5 \right]_0^1$$

不定积分, 不用写 “+C”

$$= \frac{1}{5} \left(\frac{1}{2} \right)^5$$

$$= \frac{1}{160}$$

d) 已知 $\int_{-1}^2 (ax^2 + 2x - 1)dx = b$ 且 $\int_1^3 (ax^2 + b)dx = 44$, 求实数 a, b 的值。

$$\int_{-1}^2 (ax^2 + 2x - 1)dx = b$$

$$\left[\frac{ax^3}{3} + x^2 - x \right]_{-1}^2 = b$$

$$\left(\frac{8a}{3} + 4 - 2 \right) - \left(-\frac{a}{3} + 1 - (-1) \right) = b$$

$$3a = b$$

$$\int_1^3 (ax^2 + b)dx = 44$$

$$\left[\frac{ax^3}{3} + bx \right]_1^3 = 44$$

$$(9a + 3b) - \left(\frac{a}{3} + b \right) = 44$$

$$\frac{26a}{3} + 2b = 44$$

解得 $a = 3, b = 9$

Chp24.2 定积分的计算

Pg. 124

不定积分:

定积分:

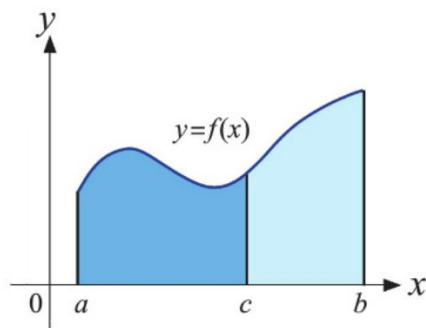
(一)提出常数法则

$$\int kf(x)dx = k \int f(x)dx, k \neq 0$$

(二)分项积分法则

$$\int f(x) \pm g(x)dx = \int f(x)dx \pm \int g(x)dx$$

由不定积分的性质, 我们可以得到性质 1 及性质 2。而利用定积分的定义, 则可得性质 3 及性质 4。当 $f(x) \geq 0$, $a \leq c \leq b$, 我们可用图像的面积来理解性质 5: 曲线 $y=f(x)$, 直线 $x=a$, $x=b$ 及 x 轴所围成的区域面积可以如图 24-6 拆分成从 a 到 c 及从 c 到 b 两个曲边梯形的面积之和。



(一)提出常数法则

$$\int_a^b kf(x)dx = k \int_a^b f(x)dx, k \text{ 为常数}$$

(二)分项积分法则

$$\int_a^b f(x) \pm g(x)dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$$

$$(三) \int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

其中 $a < c < b$

$$(四) \int_a^b f(x)dx = -\int_b^a f(x)dx$$

a) 求 $i) \int_0^1 (2x + x^3) dx$

$$\int_0^1 (2x + x^3) dx$$

$$= \left[x^2 + \frac{x^4}{4} \right]_0^1$$

$$= \left(1 + \frac{1}{4} \right)$$

$$= \frac{5}{4}$$

ii) $\int_2^3 (x - 2)(x - 3) dx$

$$\int_2^3 (x^2 - 5x + 6) dx$$

$$= \left[\frac{x^3}{3} - \frac{5x^2}{2} + 6x \right]_2^3$$

$$= \left(\frac{3^3}{3} - \frac{5(3)^2}{2} + 6(3) \right)$$

$$- \left(\frac{2^3}{3} - \frac{5(2)^2}{2} + 6(2) \right)$$

$$= -\frac{1}{6}$$

iii) $\int_0^1 x(1 - \sqrt{x})^2 dx$

$$\int_0^1 x(1 - 2\sqrt{x} + x) dx$$

$$= \int_0^1 x - 2x^{\frac{3}{2}} + x^2 dx$$

$$= \left[\frac{x^2}{2} - \frac{(2)2x^{\frac{5}{2}}}{5} + \frac{x^3}{3} \right]_0^1$$

$$= \frac{1}{2} - \frac{4}{5} + \frac{1}{3}$$

$$= \frac{1}{30}$$

a) 求iii) 已知 $f(x) = \begin{cases} 3x-1, & -1 \leq x \leq 1 \\ x+1, & 1 < x \leq 3 \end{cases}$, 求 $\int_{-1}^3 f(x) dx$

$$\int_{-1}^3 f(x) dx$$

$$= \int_{-1}^1 f(x) dx + \int_1^3 f(x) dx$$

$$= \int_{-1}^1 (3x-1) dx + \int_1^3 (x+1) dx$$

$$= \left[\frac{3x^2}{2} - x \right]_{-1}^1 + \left[\frac{x^2}{2} + x \right]_1^3$$

$$= \left(\left(\frac{3}{2} - 1 \right) - \left(\frac{3}{2} + 1 \right) \right) + \left(\left(\frac{9}{2} + 3 \right) - \left(\frac{1}{2} + 1 \right) \right)$$

$$= 4$$

iv) 已知 $f(x) = \begin{cases} x^2 - 1, & -2 \leq x < 1 \\ 1 - x, & 1 \leq x \leq 2 \end{cases}$, 求 $\int_{-2}^2 f(x) dx$

$$\int_{-2}^2 f(x) dx$$

$$= \int_{-2}^1 f(x) dx + \int_1^2 f(x) dx$$

$$= \int_{-2}^1 (x^2 - 1) dx + \int_1^2 (1 - x) dx$$

$$= \left[\frac{x^3}{3} - x \right]_{-2}^1 + \left[x - \frac{x^2}{2} \right]_1^2$$

$$= \left(\left(\frac{1}{3} - 1 \right) - \left(-\frac{8}{3} + 2 \right) \right) + \left((2 - 2) - \left(1 - \frac{1}{2} \right) \right)$$

$$= -\frac{1}{2}$$

换元积分法

若 g' 在区间 $[a, b]$ 上连续且 f 在 $u = g(x)$ 的值域上连续, 则

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$

证: 令 F 为 f 的原函数, 即 $\frac{d}{dx}F(x) = f(x)$, 由微积分基本定理第二部分可得,

$$\begin{aligned}\int_{g(a)}^{g(b)} f(u)du &= [F(u)]_{g(a)}^{g(b)} \\ &= F(g(b)) - F(g(a))\end{aligned}$$

另外, 因为 $\frac{d}{dx}F(g(x)) = F'(g(x))g'(x) = f(g(x))g'(x)$, 所以 $F(g(x))$ 是 $f(g(x))g'(x)$ 的原函数。同样, 由微积分基本定理第二部分可得,

$$\begin{aligned}\int_a^b f(g(x))g'(x)dx &= [F(g(x))]_a^b \\ &= F(g(b)) - F(g(a))\end{aligned}$$

由此可得, $\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$ 。

$$b) \text{求} i) \int_0^3 \frac{1}{\sqrt{1+x}} dx$$

$$\text{设 } u = 1 + x$$

$$\frac{du}{dx} = 1$$

$$\text{当 } x = 3, u = 4$$

$$\text{当 } x = 0, u = 1$$

$$du = dx$$

$$\begin{aligned} \int_0^3 \frac{1}{\sqrt{1+x}} dx &= \int_1^4 \frac{1}{\sqrt{u}} du \\ &= \int_1^4 u^{-\frac{1}{2}} du \\ &= \left[2u^{\frac{1}{2}} \right]_1^4 \\ &= 2 \left[\sqrt{u} \right]_1^4 \\ &= 2[2 - 1] \\ &= 2 \end{aligned}$$



注意

在利用 $u = g(x)$ 作代换时，我们必须将积分变量 x 的上、下限也换成相对应的 u 的上、下限。

$$ii) \int_1^9 \frac{x+1}{\sqrt{x}} dx$$

$$\begin{aligned} &\int_1^9 \frac{x+1}{\sqrt{x}} dx \\ &= \int_1^9 \frac{x}{\sqrt{x}} + \frac{1}{\sqrt{x}} dx \\ &= \int_1^9 x^{\frac{1}{2}} + x^{-\frac{1}{2}} dx \\ &= \left[\frac{2x^{\frac{3}{2}}}{3} + 2x^{\frac{1}{2}} \right]_1^9 \end{aligned}$$

分母一项，拆分

$$\begin{aligned} &= \left[\frac{2\sqrt{x^3}}{3} + 2\sqrt{x} \right]_1^9 \\ &= \left(\frac{54}{3} + 6 \right) - \left(\frac{2}{3} + 2 \right) \\ &= \frac{64}{3} \end{aligned}$$

b)求iii) $\int_0^2 \frac{x}{(4+x)^3} dx$

设 $u = 4 + x \Rightarrow x = u - 4$ 当 $x = 2, u = 6$

$$\frac{du}{dx} = 1$$

当 $x = 0, u = 4$

$$du = dx$$

$$\int_0^2 \frac{x}{(4+x)^3} dx = \int_4^6 \frac{u-4}{u^3} du$$

分母一项，拆分

$$= \int_4^6 u^{-2} - 4u^{-3} du$$

$$= \left[\frac{u^{-1}}{-1} - \frac{4u^{-2}}{-2} \right]_4^6$$

$$= \left[-\frac{1}{u} + \frac{2}{u^2} \right]_4^6$$

$$= \left(-\frac{1}{6} + \frac{2}{6^2} \right) - \left(-\frac{1}{4} + \frac{2}{4^2} \right)$$

$$= \frac{1}{72}$$



注意

在利用 $u = g(x)$ 作代换时，我们必须将积分变量 x 的上、下限也换成相对应的 u 的上、下限。

c) 已知 $\int_0^2 f(x)dx = 7, \int_6^2 f(x)dx = 9$, 求

i) $\int_2^6 f(x)dx$

$$\int_2^6 f(x)dx$$

$$= -\int_6^2 f(x)dx$$

$$= -9$$

ii) $\int_0^6 f(x)dx$

$$\int_0^6 f(x)dx$$

$$= \int_0^2 f(x)dx + \int_2^6 f(x)dx$$

$$= 7 + (-9)$$

$$= -2$$

iii) $\int_0^6 [3f(x) + x^2]dx$

$$\int_0^6 [3f(x) + x^2]dx$$

$$= 3\int_0^6 f(x)dx + \int_0^6 x^2 dx$$

$$= 3(-2) + \left[\frac{x^3}{3} \right]_0^6$$

$$= -6 + \frac{6^3}{3}$$

$$= 66$$

$$\int_a^b kf(x)dx = k \int_a^b f(x)dx$$

$$\int_a^b f(x) \pm g(x)dx$$

$$= \int_a^b f(x)dx \pm \int_a^b g(x)dx$$

$$\int_a^b f(x)dx$$

$$= \int_a^c f(x)dx + \int_c^b f(x)dx$$

$$\int_a^b f(x)dx = -\int_b^a f(x)dx$$

c) 已知 $\int_0^2 f(x)dx = 7$, $\int_6^2 f(x)dx = 9$, 求

$$iv) \int_0^1 f(2x)dx$$

设 $u = 2x$ 当 $x = 1, u = 2$

$\frac{du}{dx} = 2$ 当 $x = 0, u = 0$

$$du = 2dx$$

$$\int_0^1 \frac{2f(2x)}{2} dx = \frac{1}{2} \int_0^2 f(u)du$$

$$\because \int_0^2 f(x)dx = 7$$

$$\therefore \int_0^2 f(\Delta)d\Delta = 7$$

$$= \frac{1}{2} \int_0^2 f(x)dx$$

$$= \frac{7}{2}$$

$$v) \int_0^2 [f(3x) + 2]dx$$

设 $u = 3x$ 当 $x = 2, u = 6$

$\frac{du}{dx} = 3$ 当 $x = 0, u = 0$

$$du = 3dx$$

$$\int_0^2 \frac{3f(3x)}{3} dx + \int_0^2 2dx = \frac{1}{3} \int_0^6 f(u)du + [2x]_0^2$$

$$= \frac{1}{3} \int_0^6 f(x)dx + 4$$

$$= \frac{1}{3}(-2) + 4 = \frac{10}{3}$$



注意

在利用 $u = g(x)$ 作代换时, 我们必须将积分变量 x 的上、下限也换成相对应的 u 的上、下限。

$$\int_a^b kf(x)dx = k \int_a^b f(x)dx$$

$$\int_a^b f(x) \pm g(x)dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$$

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

$$\int_a^b f(x)dx = -\int_b^a f(x)dx$$

c) 已知 $\int_0^2 f(x)dx = 7$, $\int_6^2 f(x)dx = 9$, 求

vi) $\int_0^1 [(4x-1)f(12x^2-6x)]dx$

设 $u = 12x^2 - 6x$ 当 $x = 1, u = 6$

$\frac{du}{dx} = 24x - 6$ 当 $x = 0, u = 0$

$du = (24x - 6)dx$

$\int_0^1 \frac{6(4x-1)}{6} f(12x^2-6x)dx = \frac{1}{6} \int_0^6 f(u)du$

$= \frac{1}{6} \int_0^6 f(x)dx$

$= \frac{1}{6}(-2) = -\frac{1}{3}$

练习 24.2 - 3, 8, 9, 10, 11, 14

$$\int_a^b kf(x)dx = k \int_a^b f(x)dx$$

$$\begin{aligned} \int_a^b f(x) \pm g(x)dx \\ = \int_a^b f(x)dx \pm \int_a^b g(x)dx \end{aligned}$$

$$\begin{aligned} \int_a^b f(x)dx \\ = \int_a^c f(x)dx + \int_c^b f(x)dx \end{aligned}$$

$$\int_a^b f(x)dx = -\int_b^a f(x)dx$$

d) 设 $f(x)$ 在区间 $[-a, a]$ 上是连续的, 利用奇、偶函数性质证明:

i) 若 f 是奇函数, 即 $f(-x) = -f(x)$, 证明 $\int_{-a}^a f(x) dx = 0$ 。

ii) 若 f 是偶函数, 即 $f(-x) = f(x)$, 证明 $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ 。

$$i) \int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

$$\text{设 } u = -x \Rightarrow x = -u$$

$$\frac{du}{dx} = -1 \quad \text{当 } x=0, u=0$$

$$du = -dx \quad \text{当 } x=-a, u=a$$

$$\text{原式} = -\int_a^0 f(-u) du + \int_0^a f(x) dx$$

$$= \int_0^a f(-x) dx + \int_0^a f(x) dx$$

$$= -\int_0^a f(x) dx + \int_0^a f(x) dx$$

$$= 0$$

$$ii) \int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

$$\text{设 } u = -x \Rightarrow x = -u$$

$$\frac{du}{dx} = -1 \quad \text{当 } x=0, u=0$$

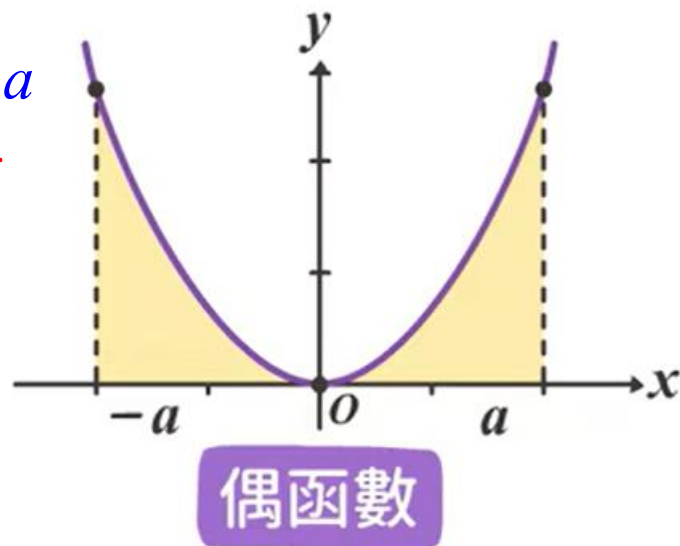
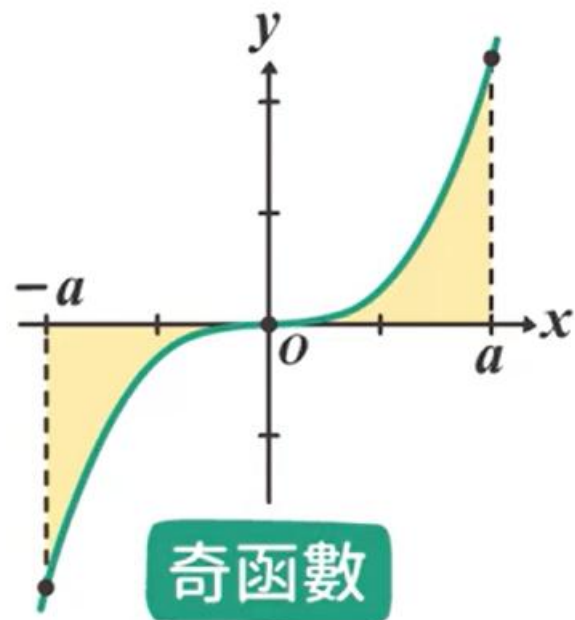
$$du = -dx \quad \text{当 } x=-a, u=a$$

$$\text{原式} = -\int_a^0 f(-u) du + \int_0^a f(x) dx$$

$$= \int_0^a f(-x) dx + \int_0^a f(x) dx$$

$$= \int_0^a f(x) dx + \int_0^a f(x) dx$$

$$= 2 \int_0^a f(x) dx$$



e) 利用奇、偶函数性质、定积分及面积概念完成下列填空：

$$\text{已知 } \int_0^5 x^3 dx = \frac{625}{4}; \int_0^5 x^4 dx = 625; \int_0^1 \left(\sqrt[5]{x} - \frac{6}{\sqrt[3]{x}} \right) dx = -\frac{49}{6}; \int_0^{\frac{\pi}{4}} \frac{x}{x^2 + 1} dx = \frac{1}{2} \ln \left(\frac{\pi^2 + 16}{16} \right);$$

$$i) \int_{-5}^5 x^3 dx = \underline{\hspace{2cm}}$$

$$ii) \int_{-5}^5 x^4 dx = \underline{\hspace{2cm}}$$

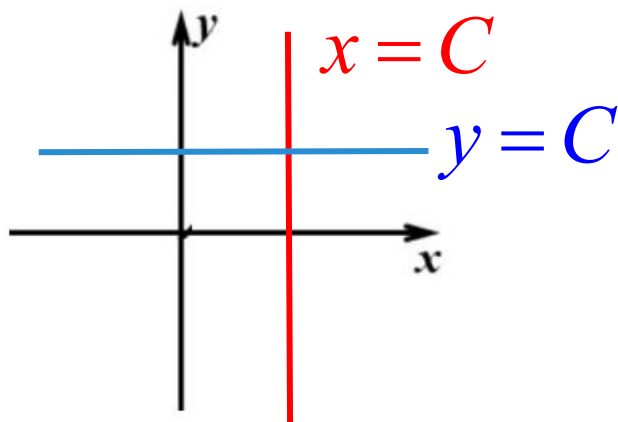
$$iii) \int_{-1}^1 \left(\sqrt[5]{x} - \frac{6}{\sqrt[3]{x}} \right) dx = \underline{\hspace{2cm}}$$

$$iv) \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x}{x^2 + 1} dx = \underline{\hspace{2cm}}$$

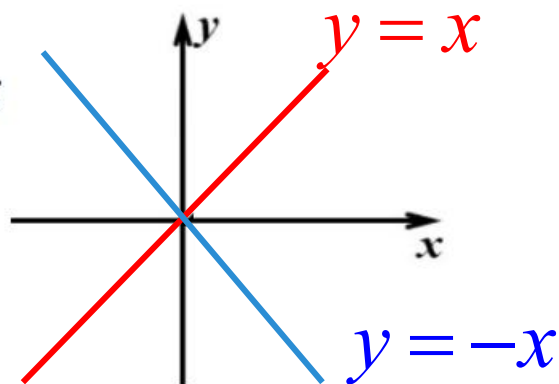
Chp24.3 面积计算 Pg. 132

几种基本的函数图像

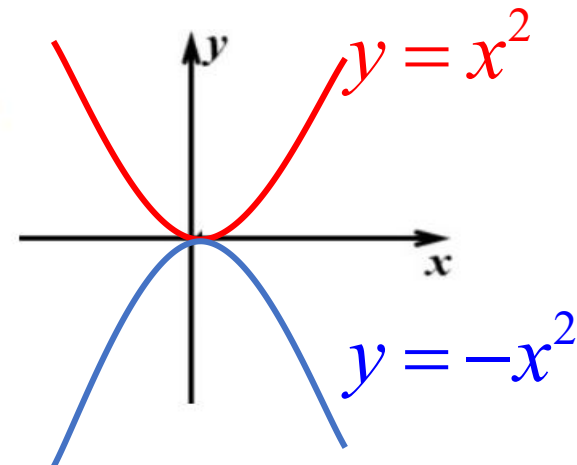
$$y = c$$
$$x = c$$



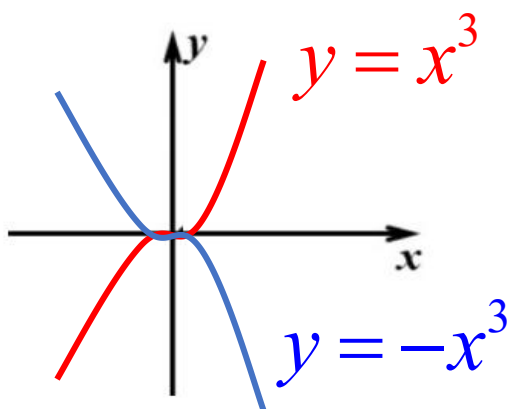
$$y = x$$
$$y = -x$$



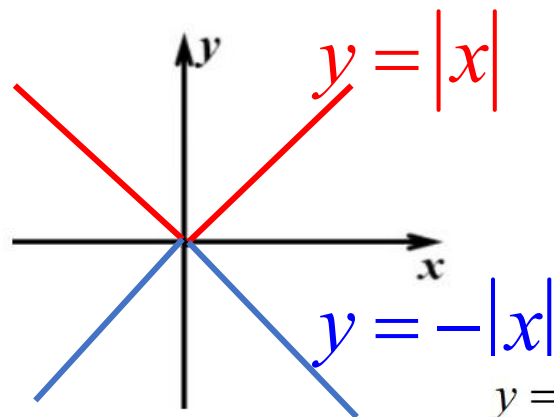
$$y = x^2$$
$$y = -x^2$$



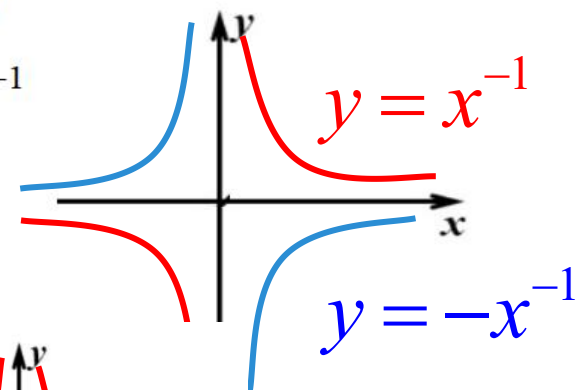
$$y = x^3$$
$$y = -x^3$$



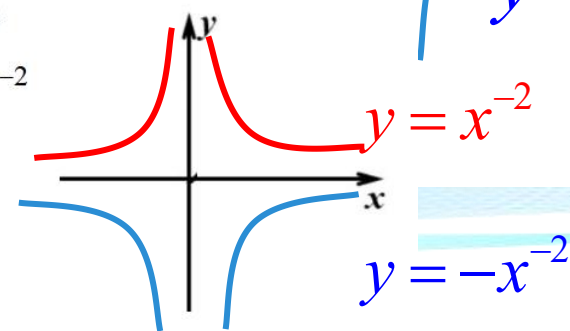
$$y = |x|$$
$$y = -|x|$$



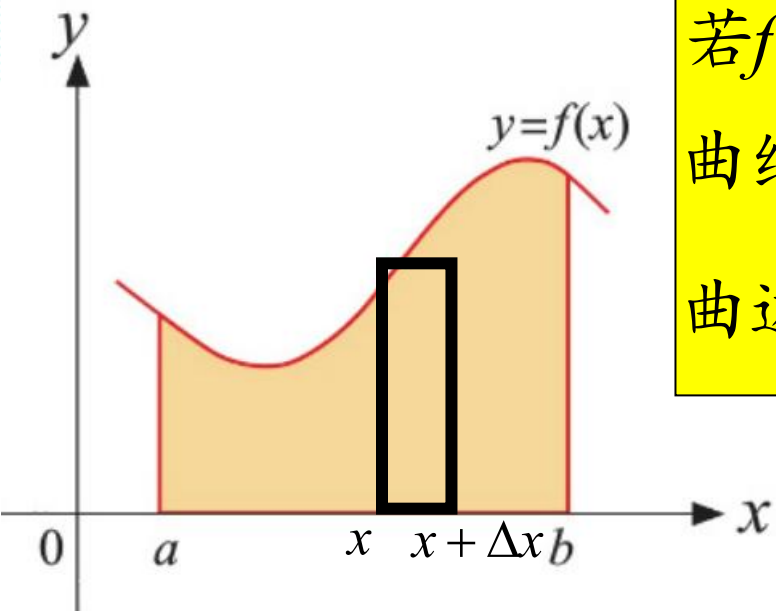
$$y = x^{-1}$$
$$y = -x^{-1}$$



$$y = x^{-2}$$
$$y = -x^{-2}$$



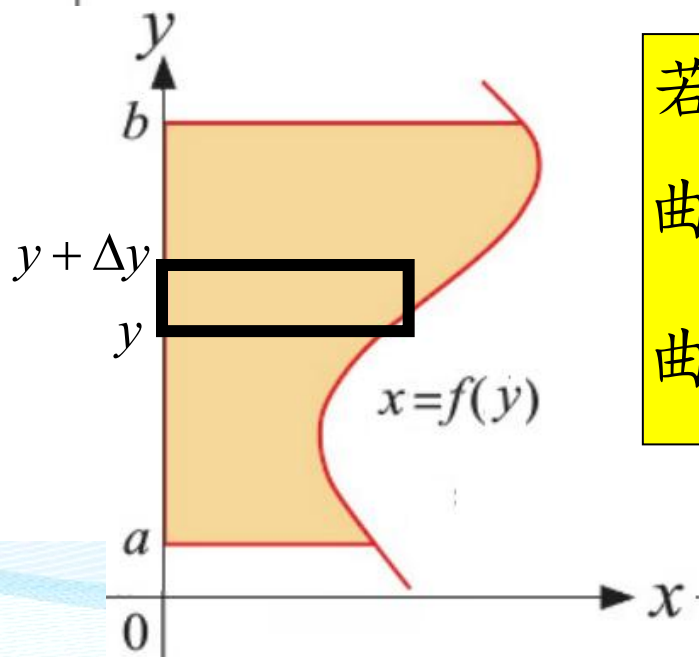
曲线与x, y轴围成的面积面积公式



若 $f(x) > 0$

曲线 $y = f(x)$ 与 x 轴, $x = a$, $x = b$ 所围成的曲边梯形的面积为 $A = \int_a^b f(x) dx$

记忆法: $A = \int_a^b \underbrace{y}_{\text{长}} \underbrace{dx}_{\text{宽}}$

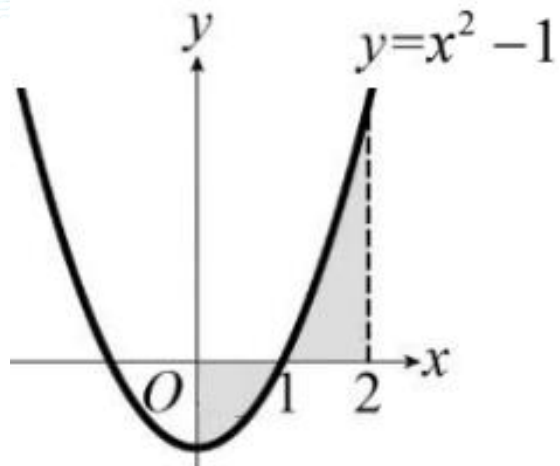


若 $f(y) > 0$

曲线 $x = f(y)$ 与 y 轴, $y = a$, $y = b$ 所围成的曲边梯形的面积为 $A = \int_a^b f(y) dy$

记忆法: $A = \int_a^b \underbrace{x}_{\text{长}} \underbrace{dy}_{\text{宽}}$

a) 如下图，求曲线 $y = x^2 - 1$ 与 $x = 0$, $x = 2$ 及 x 轴所围成的面积。



常犯错误

$$A = \int_0^2 (x^2 - 1) dx$$

$$= \left[\frac{x^3}{3} - x \right]_0^2 = \frac{8}{3} - 2 = \frac{2}{3}$$

$$A = \left| \int_0^1 (x^2 - 1) dx \right| + \int_1^2 (x^2 - 1) dx$$

$$= \left[\frac{x^3}{3} - x \right]_0^1 + \left[\frac{x^3}{3} - x \right]_1^2$$

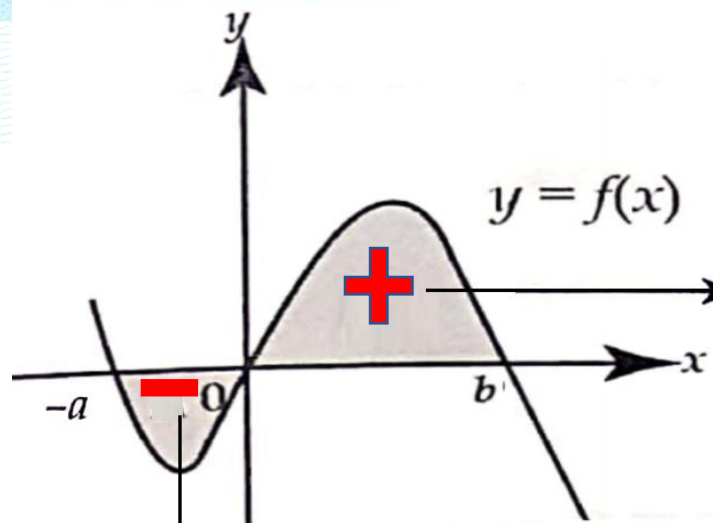
$$= \left| \frac{1}{3} - 1 \right| + \left(\left(\frac{8}{3} - 2 \right) - \left(\frac{1}{3} - 1 \right) \right)$$

$$= \left| -\frac{2}{3} \right| + \frac{4}{3}$$

$$= 2 \text{ (平方单位)}$$

阴影部分在 x 轴上方

其 $\int_0^b f(x)dx$ 为正值



$\therefore$ 阴影部分面积 $= \int_0^b f(x)dx$

阴影部分在 x 轴下方

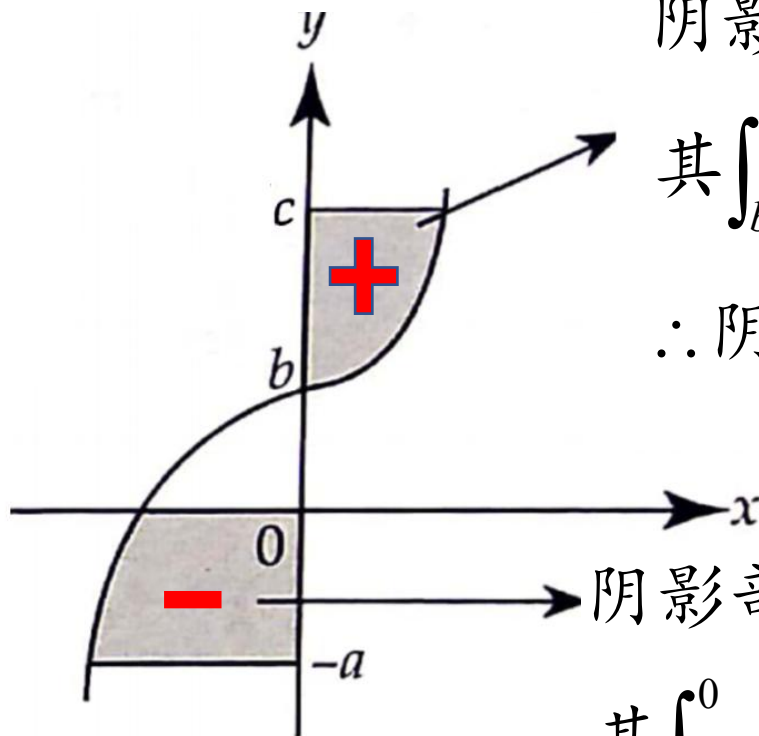
其 $\int_{-a}^0 f(x)dx$ 为负值

$\therefore$ 阴影部分面积 $= \left| \int_{-a}^0 f(x)dx \right|$

阴影部分在 y 轴右方

其 $\int_b^c f(y)dy$ 为正值

$\therefore$ 阴影部分面积 $= \int_b^c f(y)dy$



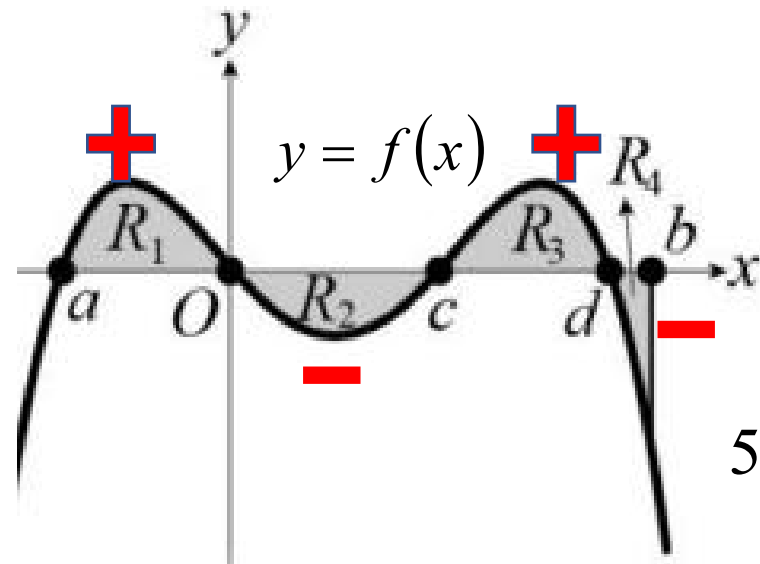
阴影部分在 y 轴左方

其 $\int_{-a}^0 f(y)dy$ 为负值

$\therefore$ 阴影部分面积 $= \left| \int_{-a}^0 f(y)dy \right|$

b)是非题

如图, 已知图纸四个区域 R_1, R_2, R_3, R_4 的面积分别为6, 8, 4, 5:



1. (✓) $\int_a^0 f(x)dx = 6$

2. (✗) $\int_0^c f(x)dx = 8$

3. (✗) $\int_d^c f(x)dx = 4$

4. (✗) $\int_a^b f(x)dx = 6 - 8 + 4 + 5$

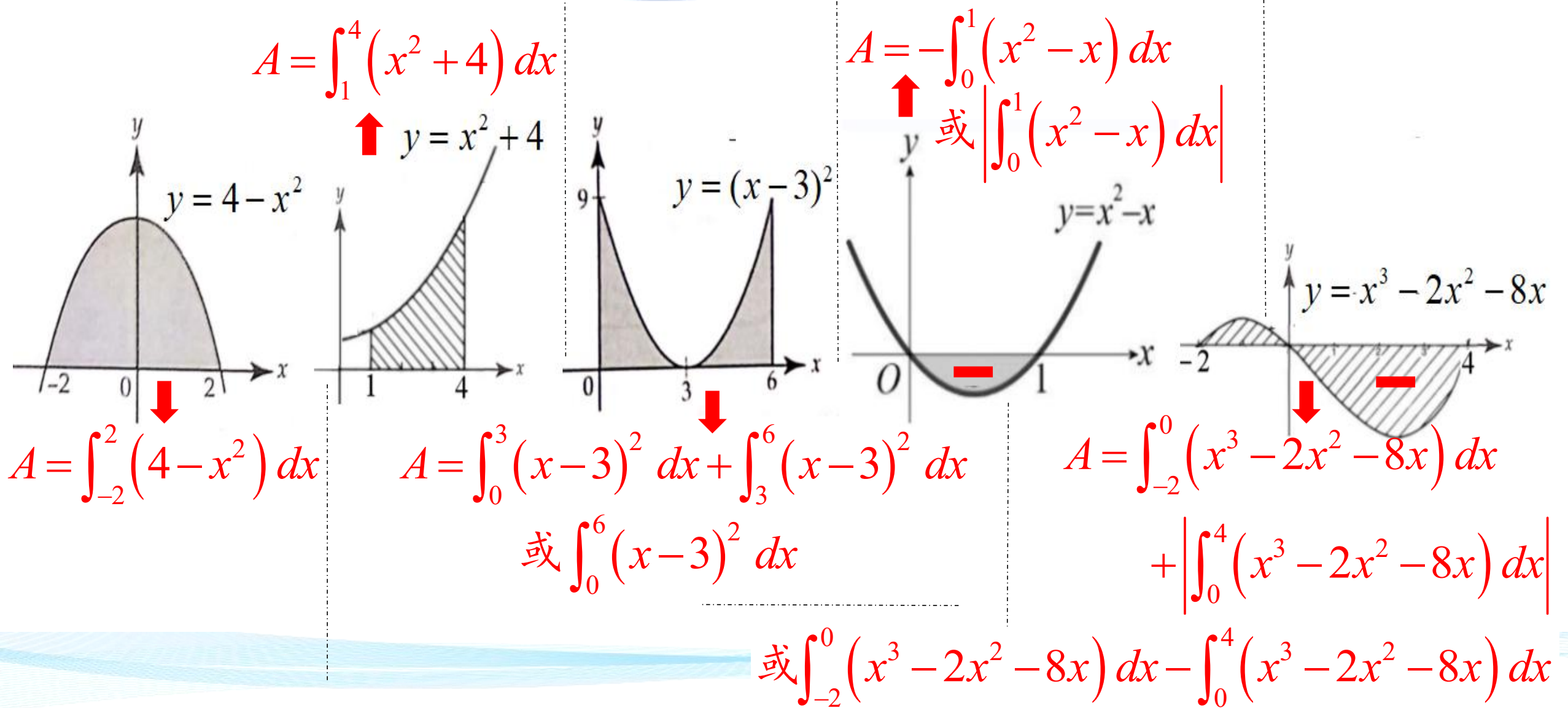
5. (✗) $\int_a^b f(x)dx = \text{阴影部分面积}$

6. (✓) $\int_a^0 f(x)dx + \left| \int_0^c f(x)dx \right| + \int_c^d f(x)dx + \left| \int_d^b f(x)dx \right| = \text{阴影部分面积}$

7. (✗) $\int_a^b f(x)dx = \left| \int_a^b f(x)dx \right|$

8. (✗) $\int_0^d f(x)dx = 8 + 4$

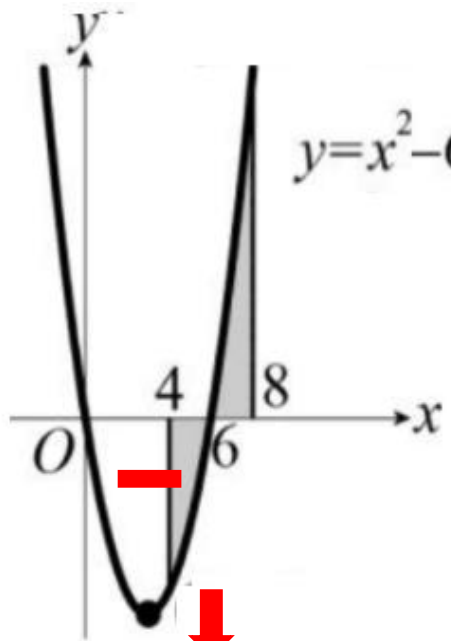
c) 用定积分表示下列图中各个阴影部分面积:



c) 用定积分表示下列图中各个阴影部分面积:

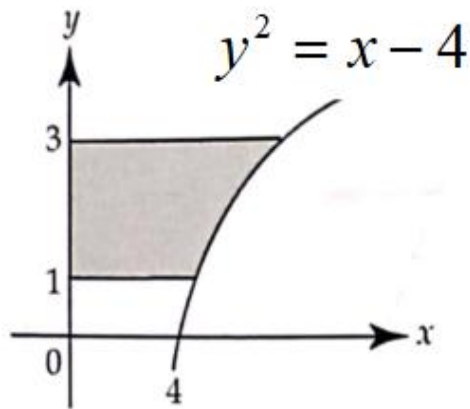
$$y^2 = x - 4 \Rightarrow x = y^2 + 4$$

$$A = \int_1^3 (y^2 + 4) dy$$



$$A = \left| \int_4^6 (x^2 - 6x) dx \right| + \int_6^8 (x^2 - 6x) dx$$

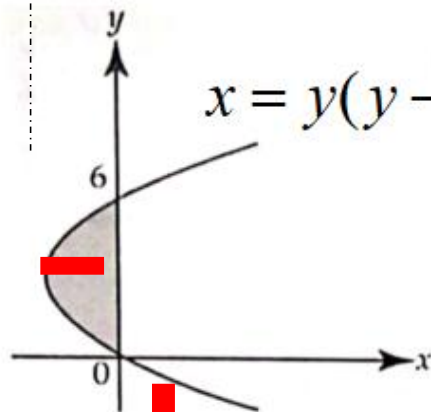
$$\text{或} -\int_4^6 (x^2 - 6x) dx + \int_6^8 (x^2 - 6x) dx$$



$$y^2 = -x \Rightarrow x = -y^2$$

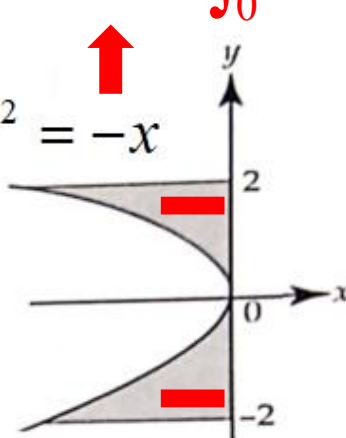
$$A = \left| \int_{-2}^2 -y^2 dy \right|$$

$$\text{或} -\int_{-2}^0 -y^2 dy - \int_0^2 -y^2 dy$$



$$A = \left| \int_0^6 y(y - 6) dy \right|$$

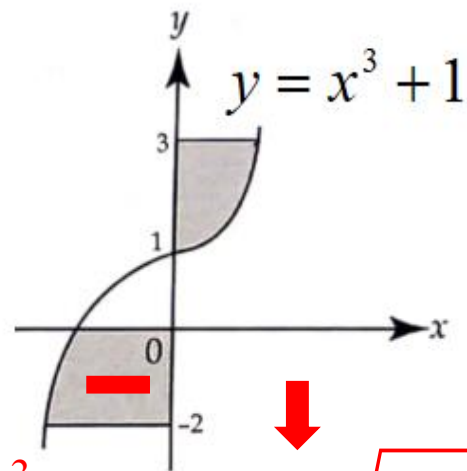
$$\text{或} -\int_0^6 y(y - 6) dy$$



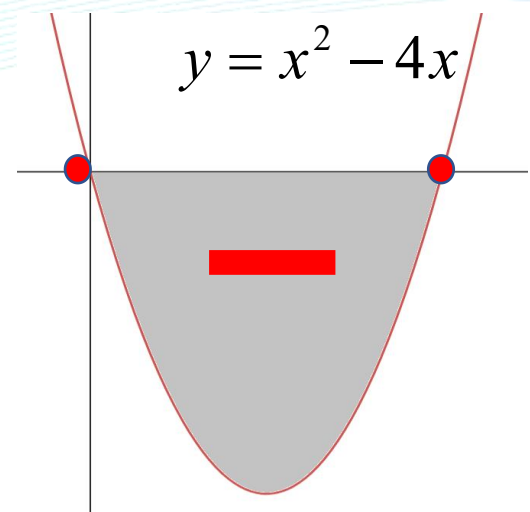
$$y = x^3 + 1 \Rightarrow x = \sqrt[3]{y - 1}$$

$$A = \left| \int_{-2}^0 \sqrt[3]{y - 1} dy \right| + \int_1^3 \sqrt[3]{y - 1} dy$$

$$\text{或} -\int_{-2}^0 \sqrt[3]{y - 1} dy + \int_1^3 \sqrt[3]{y - 1} dy$$



d)求曲线 $y = x^2 - 4x$ 与 x 轴所围成的面积。e)求曲线 $y^2 = x$, 直线 $y = 3$ 及 y 轴所围成区域的面积。



找交点 $\Leftrightarrow$ 代入法

$$\begin{cases} y = x^2 - 4x & \text{---(1)} \end{cases}$$

$$\begin{cases} y = 0 & \text{---(2)} \end{cases}$$

(2)代入(1), $x(x-4) = 0$

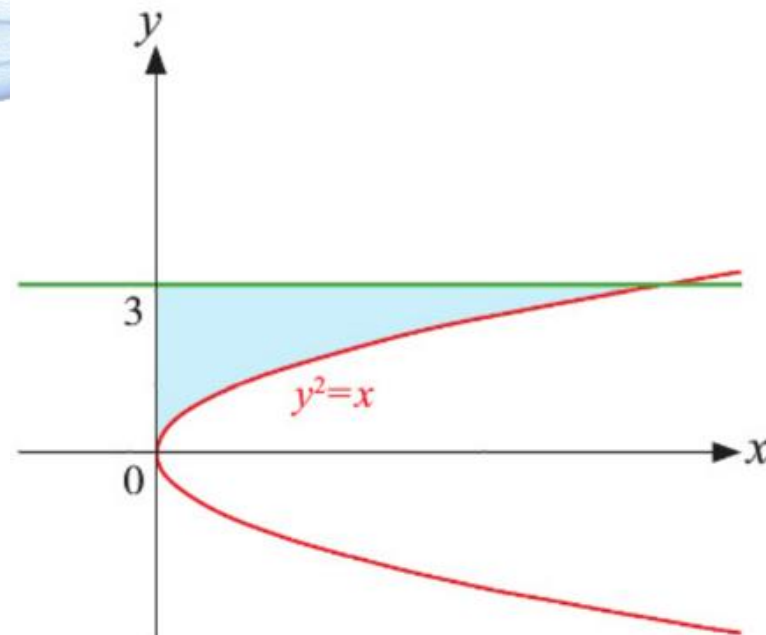
$$x = 0 \quad \text{或} \quad x = 4$$

$$A = \left| \int_0^4 (x^2 - 4x) dx \right|$$

$$= \left| \left[\frac{x^3}{3} - 2x^2 \right]_0^4 \right|$$

$$= \left| \left(\frac{64}{3} - 32 \right) - 0 \right|$$

$$= \frac{32}{3} \quad (\text{平方单位})$$



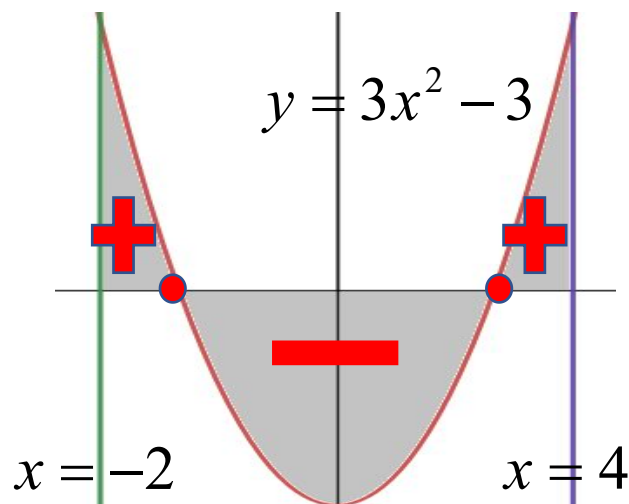
$$A = \int_0^3 y^2 dy$$

$$= \left[\frac{y^3}{3} \right]_0^3$$

$$= 9 - 0$$

$$= 9 \quad (\text{平方单位})$$

f) 求曲线 $y = 3x^2 - 3$, $x = -2$, $x = 4$ 及 x 轴所围成的面积。



找交点 $\Leftrightarrow$ 代入法

$$\begin{cases} y = 3x^2 - 3 & (1) \\ y = 0 & (2) \end{cases}$$

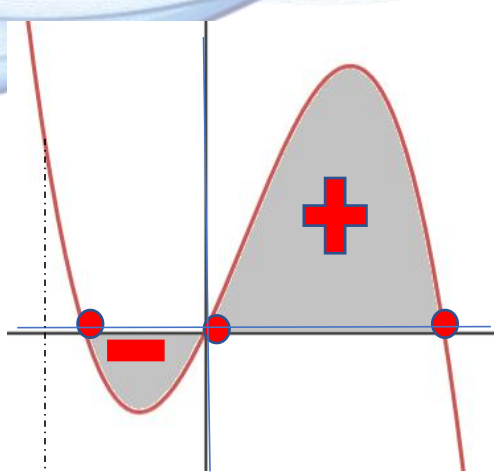
(2) 代入 (1), $3(x^2 - 1) = 0$
 $x = 1$ 或 $x = -1$

$$\begin{aligned} A &= \int_{-2}^{-1} (3x^2 - 3) dx + \left| \int_{-1}^1 (3x^2 - 3) dx \right| + \int_1^4 (3x^2 - 3) dx \\ &= \left[x^3 - 3x \right]_{-2}^{-1} + \left| \left[x^3 - 3x \right]_{-1}^1 \right| + \left[x^3 - 3x \right]_1^4 \\ &= 4 + |-4| + 54 \\ &= 62 \quad (\text{平方单位}) \end{aligned}$$

g) 求曲线 $y = -x(x-2)(x+1)$ 与 x 轴所围成的面积。

找交点 $\Leftrightarrow$ 代入法

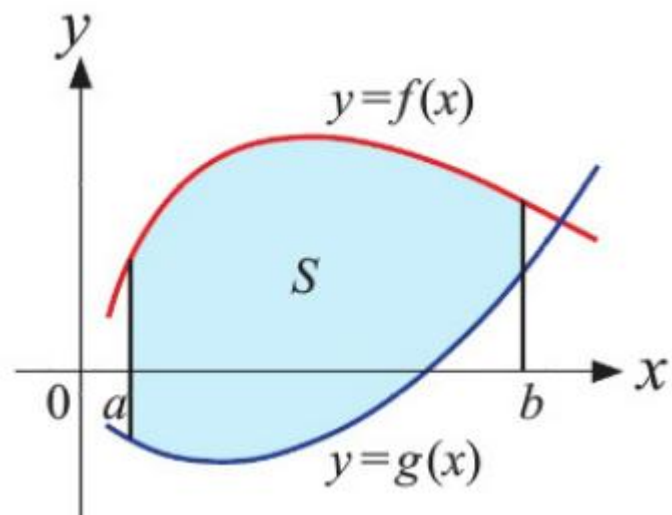
令 $-x(x-2)(x+1) = 0$
 $x = 0$ 或 $x = 2$ 或 $x = -1$



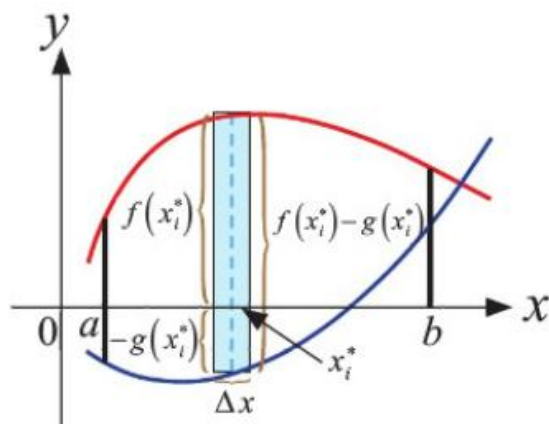
$$\begin{aligned} A &= \left| \int_{-1}^0 (-x^3 + x^2 + 2x) dx \right| + \int_0^2 (-x^3 + x^2 + 2x) dx \\ &= \left| \left[-\frac{x^4}{4} + \frac{x^3}{3} + x^2 \right]_{-1}^0 \right| + \left[-\frac{x^4}{4} + \frac{x^3}{3} + x^2 \right]_0^2 \\ &= \left| -\frac{5}{12} \right| + \frac{8}{3} \\ &= \frac{37}{12} \quad (\text{平方单位}) \end{aligned}$$

习题 24.3(1) - 1, 2, 4, 6

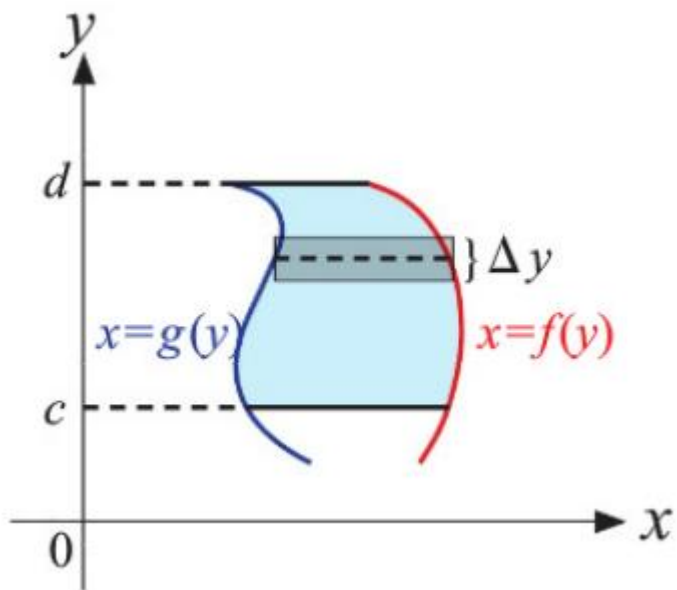
两曲线所围成的面积公式



曲线 $y=f(x)$ 与 $y=g(x)$, 直线 $x=a$, 直线 $x=b$ 所围成的区域面积为 $A = \int_a^b [f(x) - g(x)] dx$



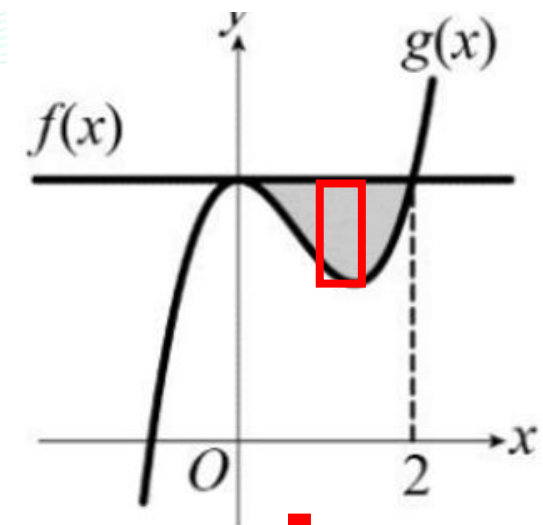
记忆法: $A = \int_a^b \underbrace{y_{\text{上}} - y_{\text{下}}}_{\text{长}} \underbrace{dx}_{\text{宽}}$



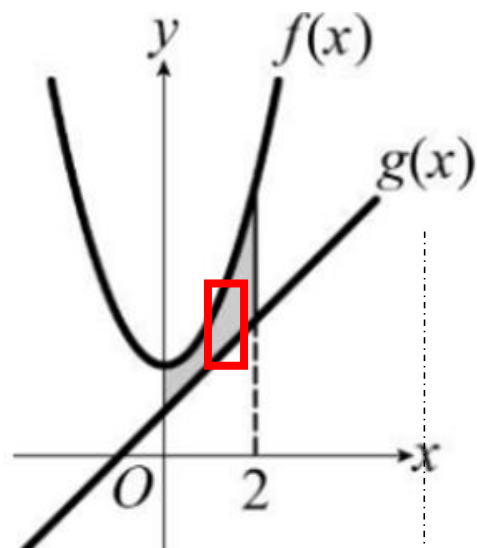
曲线 $x=f(y)$ 与 $x=g(y)$, 直线 $y=c$, 直线 $y=d$ 所围成的区域面积为 $A = \int_c^d [f(y) - g(y)] dy$

记忆法: $A = \int_c^d \underbrace{x_{\text{右}} - x_{\text{左}}}_{\text{长}} \underbrace{dy}_{\text{宽}}$

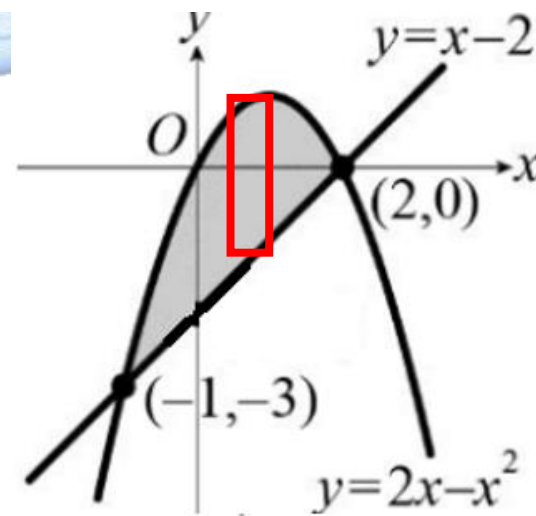
a) 用定积分表示下列图中各个阴影部分面积：



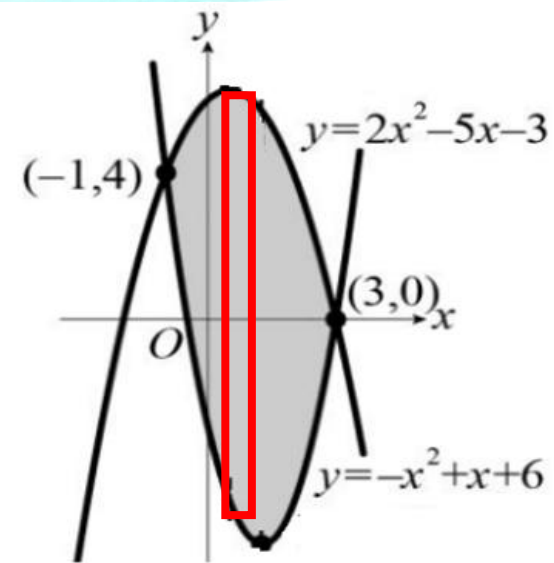
$$A = \int_0^2 [f(x) - g(x)] dx$$



$$A = \int_0^2 [f(x) - g(x)] dx$$

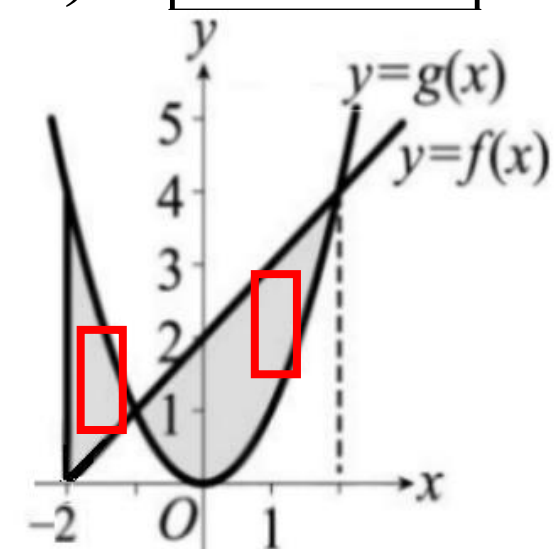


$$A = \int_{-1}^2 [(2x - x^2) - (x - 2)] dx$$

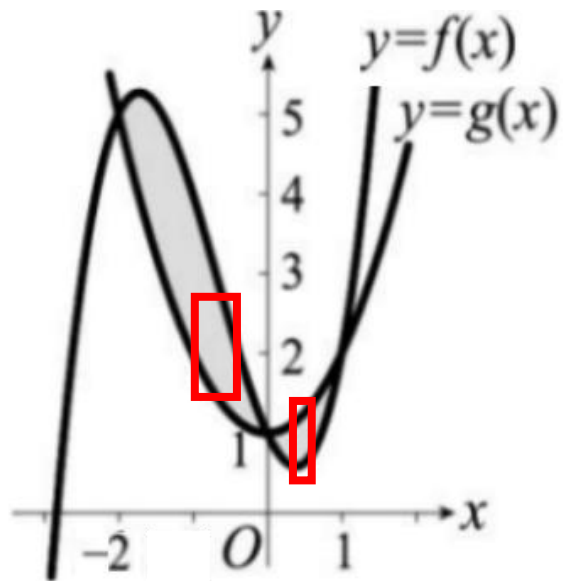


$$A = \int_{-1}^3 [(-x^2 + x + 6) - (2x^2 - 5x - 3)] dx$$

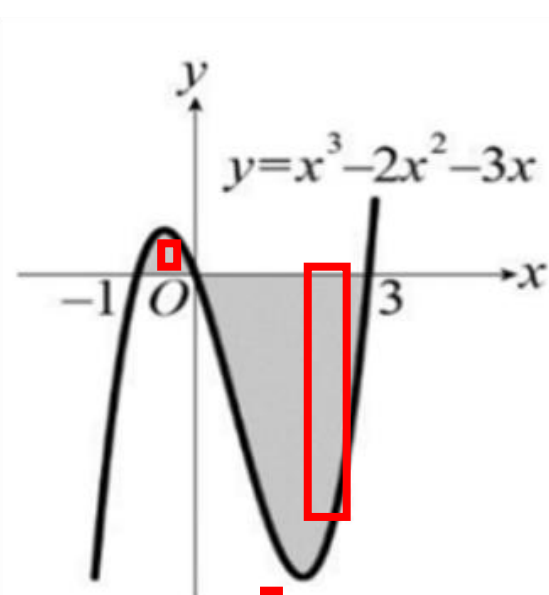
a) 用定积分表示下列图中各个阴影部分面积：



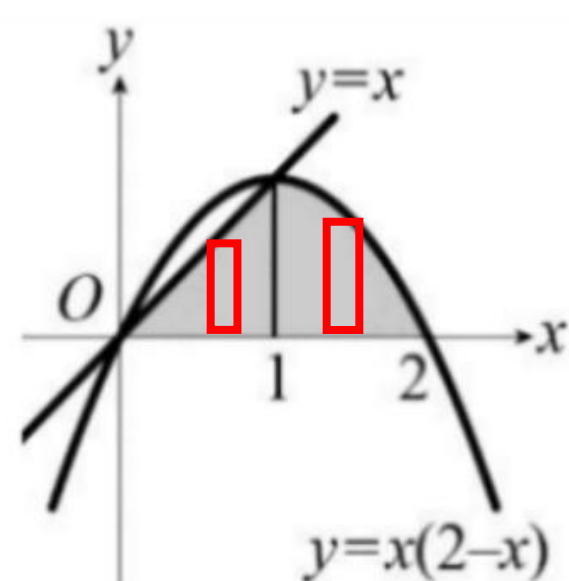
$$A = \int_{-2}^0 [g(x) - f(x)] dx + \int_{-1}^2 [f(x) - g(x)] dx$$



$$A = \int_{-2}^0 [f(x) - g(x)] dx + \int_0^1 [g(x) - f(x)] dx$$

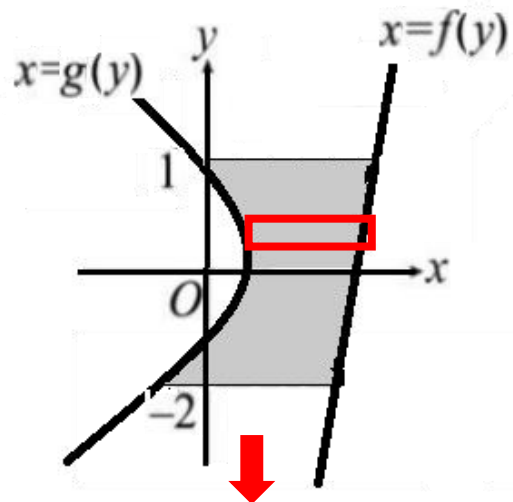


$$A = \int_{-1}^0 [(x^3 - 2x^2 - 3x) - 0] dx + \int_0^3 [0 - (x^3 - 2x^2 - 3x)] dx$$

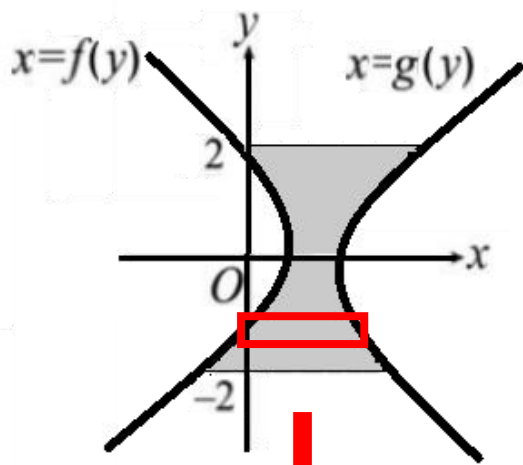


$$A = \int_0^1 [x - 0] dx + \int_1^2 [x(2-x) - 0] dx$$

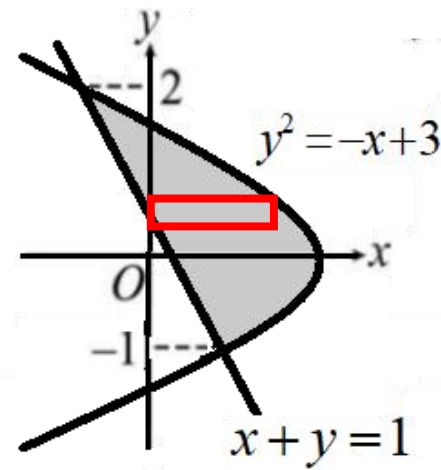
a) 用 定积分 表示下列图中各个阴影部分面积:



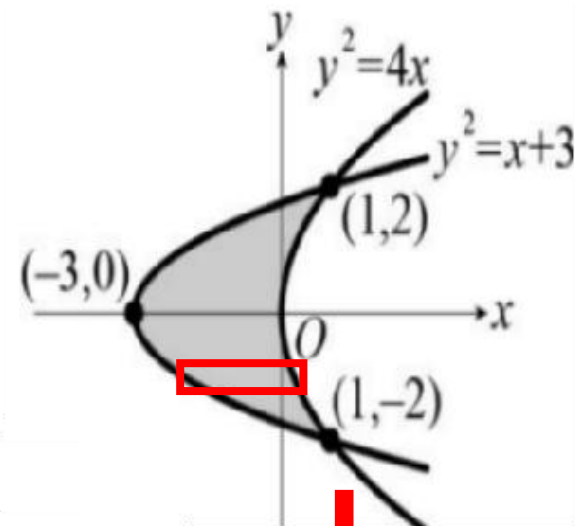
$$A = \int_{-2}^1 [f(y) - g(y)] dy$$



$$A = \int_{-2}^2 [g(y) - f(y)] dy$$

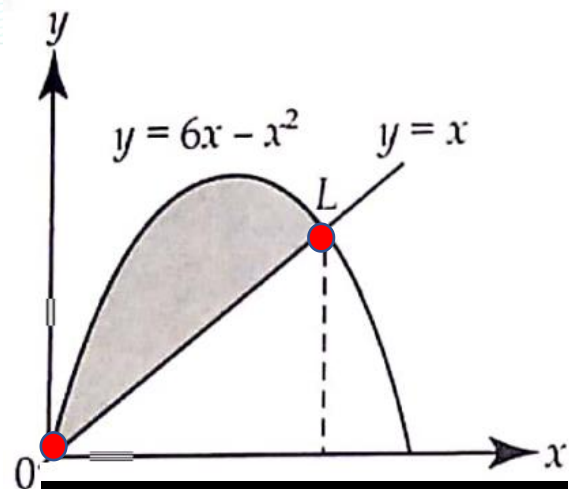


$$A = \int_{-1}^2 [(3 - y^2) - (1 - y)] dy$$



$$A = \int_{-2}^2 \left[\frac{y^2}{4} - (y^2 - 3) \right] dy$$

b)如图，求下列各个阴影部分面积：



找交点 $\Leftrightarrow$ 代入法

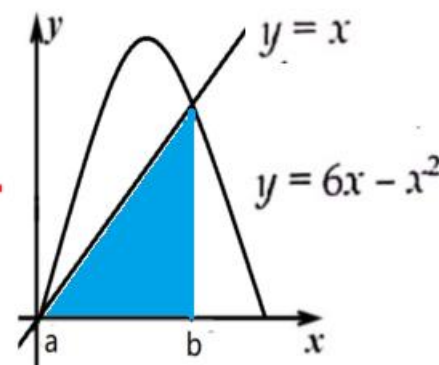
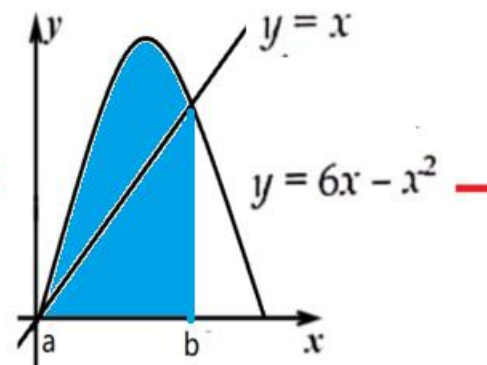
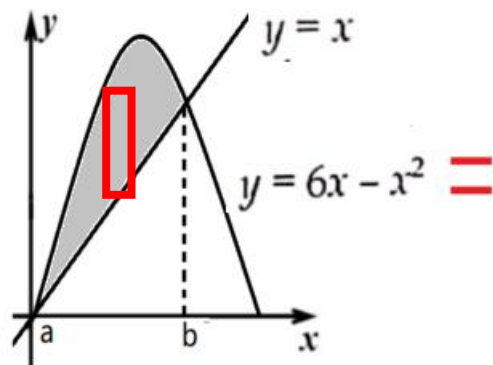
$$\begin{cases} y = 6x - x^2 & \text{---(1)} \\ y = x & \text{---(2)} \end{cases}$$

(2)代入(1), $6x - x^2 = x$

$$x^2 - 5x = 0$$

$$x(x - 5) = 0$$

$$x = 0 \quad \text{或} \quad x = 5$$



$$A = \int_0^5 (6x - x^2) dx - \int_0^5 x dx \quad \text{或}$$

$$= \left[3x^2 - \frac{x^3}{3} \right]_0^5 - \left[\frac{x^2}{2} \right]_0^5$$

$$= \left(75 - \frac{125}{3} \right) - \frac{25}{2}$$

$$= \frac{125}{6} \quad (\text{平方单位})$$

$$A = \int_0^5 \left[(6x - x^2) - (x) \right] dx$$

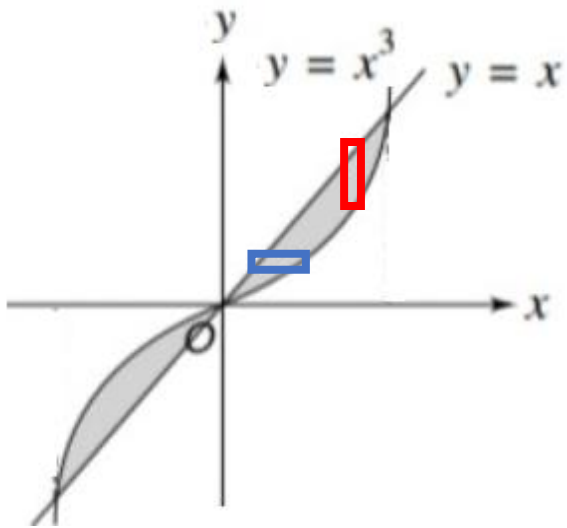
$$= \int_0^5 (5x - x^2) dx$$

$$= \left[\frac{5x^2}{2} - \frac{x^3}{3} \right]_0^5$$

$$= \frac{125}{2} - \frac{125}{3}$$

$$= \frac{125}{6} \quad (\text{平方单位})$$

b)如图, 求下列各个阴影部分面积:



解1: 对x轴

$$\begin{cases} y = x^3 & \text{--- (1)} \\ y = x & \text{--- (2)} \end{cases}$$

(2)代入(1), $x^3 = x$

$$x(x^2 - 1) = 0$$

$$x = 0 \quad \text{或} \quad x = \pm 1$$

$$A = 2 \times \int_0^1 (x - x^3) dx$$

$$= 2 \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1$$

$$= 2 \left(\frac{1}{2} - \frac{1}{4} \right)$$

$$= \frac{1}{2} \quad (\text{平方单位})$$

解2: 对y轴

$$\text{已知 } x = 0 \quad \text{或} \quad x = 1 \quad \text{或} \quad x = -1$$

$$y = 0 \quad \text{或} \quad y = 1 \quad \text{或} \quad y = -1$$

$$y = x^3 \Rightarrow x = y^{\frac{1}{3}}$$

$$A = 2 \times \int_0^1 \left(y^{\frac{1}{3}} - y \right) dy$$

$$= 2 \left[\frac{3y^{\frac{4}{3}}}{4} - \frac{y^2}{2} \right]_0^1$$

$$= 2 \left(\frac{3}{4} - \frac{1}{2} \right)$$

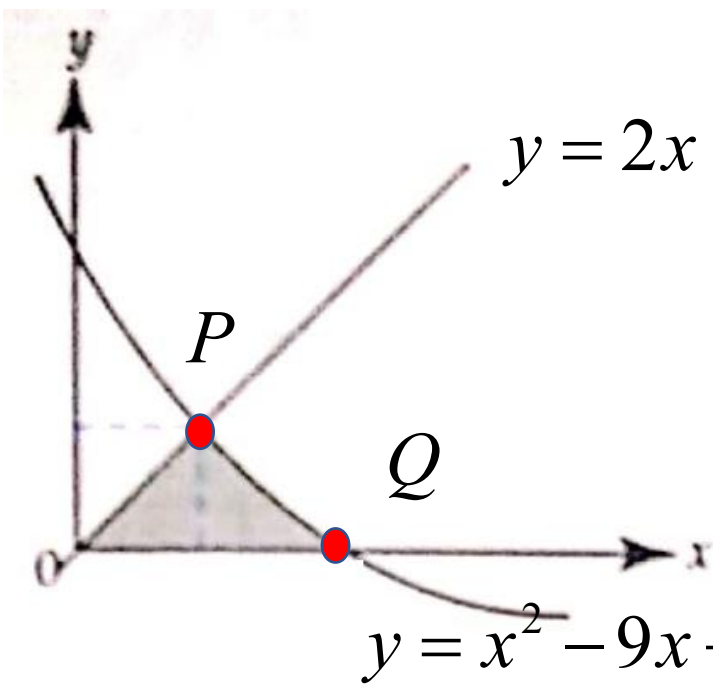
$$= \frac{1}{2} \quad (\text{平方单位})$$

c)如图, 曲线 $y = x^2 - 9x + 18$ 与 $y = 2x$ 相交于点 P , 求

i) Q 点;

ii) P 点;

iii) 阴影部分面积。



Q 为曲线与 x 轴交点

$$i) \begin{cases} y = x^2 - 9x + 18 & \text{--- (1)} \\ y = 0 & \text{--- (2)} \end{cases}$$

$$(2) \text{ 代入 (1), } x^2 - 9x + 18 = 0$$

$$(x - 3)(x - 6) = 0$$

$$x = 3 \quad \text{或} \quad x = 6 (\text{不合})$$

$$\therefore Q(3, 0)$$

P 为曲线与直线交点

$$ii) \begin{cases} y = x^2 - 9x + 18 & \text{--- (1)} \\ y = 2x & \text{--- (3)} \end{cases}$$

$$(3) \text{ 代入 (1), } x^2 - 9x + 18 = 2x$$

$$x^2 - 11x + 18 = 0$$

$$(x - 2)(x - 9) = 0$$

$$x = 2 \quad \text{或} \quad x = 9 (\text{不合})$$

$$\text{代入 (3), } y = 4$$

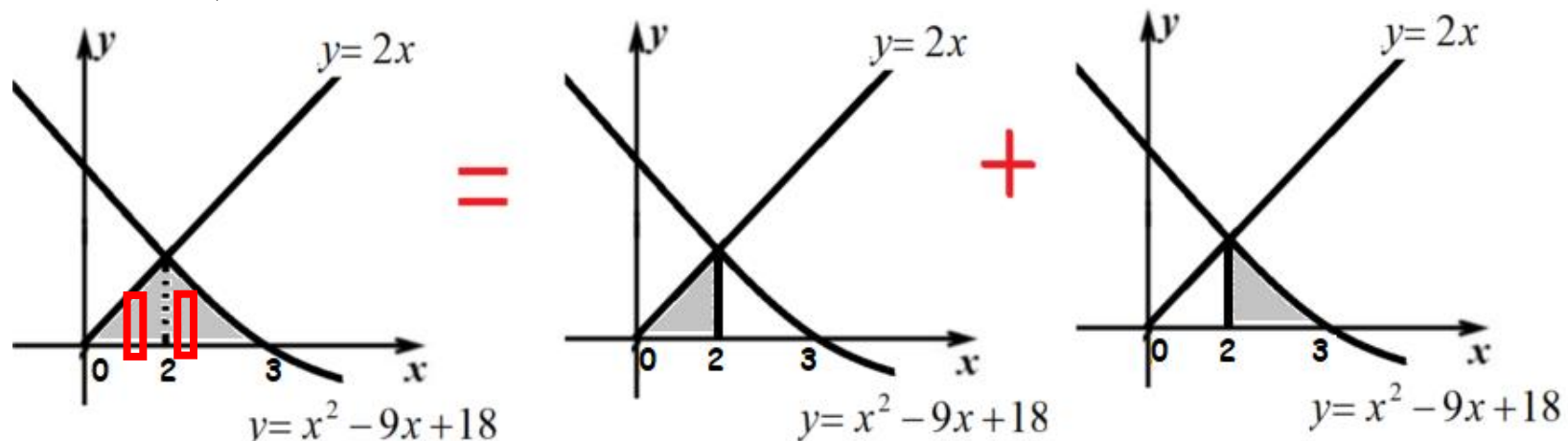
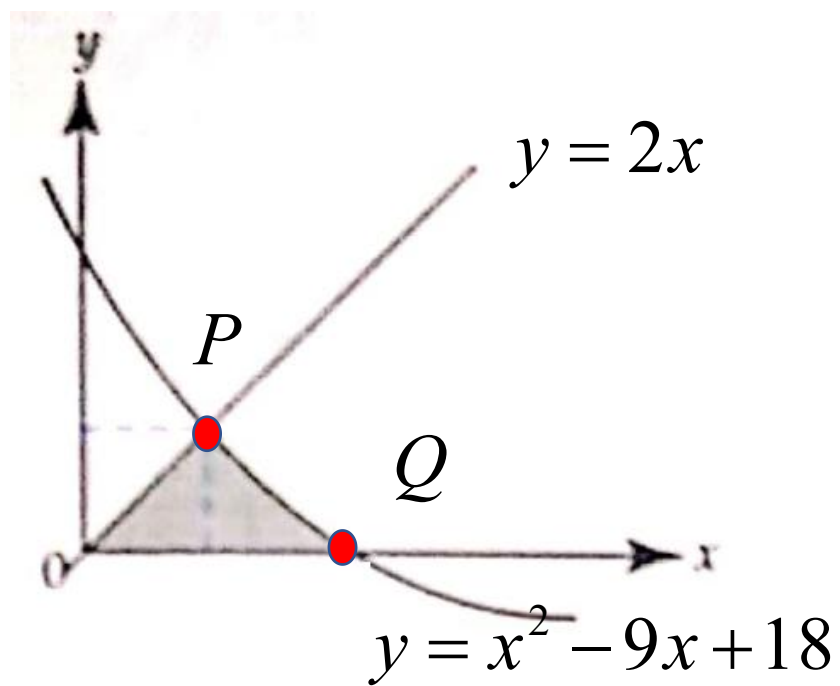
$$\therefore P(2, 4)$$

c) 如图，曲线 $y = x^2 - 9x + 18$ 与 $y = 2x$ 相交于点 P ，求

i) Q 点；

ii) P 点；

iii) 阴影部分面积。



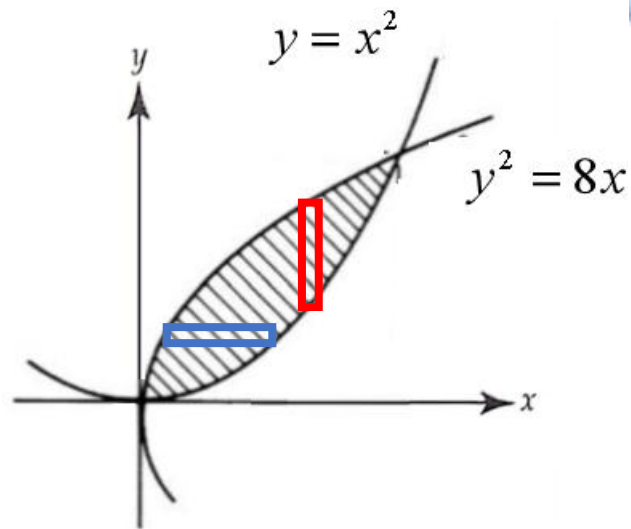
$$A = \int_0^2 2x \, dx + \int_2^3 (x^2 - 9x + 18) \, dx$$

$$= \left[x^2 \right]_0^2 + \left[\frac{x^3}{3} - \frac{9x^2}{2} + 18x \right]_2^3$$

$$= [(4) - 0] + \left[\left(9 - \frac{81}{2} + 54 \right) - \left(\frac{8}{3} - 18 + 36 \right) \right]$$

$$= \frac{35}{6} \quad (\text{平方单位})$$

d)求曲线 $y = x^2$ 与 $y^2 = 8x$ 所围成的面积。



解1:对x轴

$$\begin{cases} y = x^2 \text{———(1)} \\ y^2 = 8x \text{———(2)} \end{cases}$$

(1)代入(2), $x^4 = 8x$

$$x(x^3 - 8) = 0$$

$$x = 0 \quad \text{或} \quad x = 2$$

$$y^2 = 8x \Rightarrow y = \sqrt{8x^{\frac{1}{2}}}$$

$$A = \int_0^2 \left(\sqrt{8x^{\frac{1}{2}}} - x^2 \right) dx$$

$$= \left[\frac{2\sqrt{8}x^{\frac{3}{2}}}{3} - \frac{x^3}{3} \right]_0^2$$

$$= \frac{8}{3} \quad (\text{平方单位})$$

解2:对y轴

已知 $x = 0$ 或 $x = 2$

$y = 0$ 或 $y = 4$

$$y = x^2 \Rightarrow x = y^{\frac{1}{2}}$$

$$A = \int_0^4 \left(y^{\frac{1}{2}} - \frac{y^2}{8} \right) dy$$

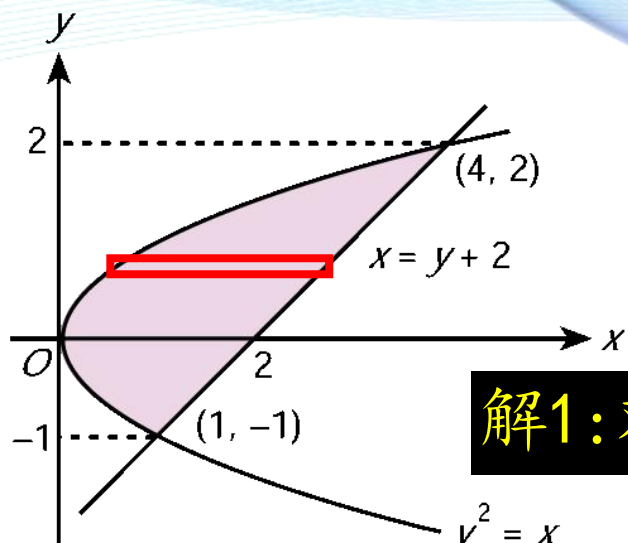
$$= \left[\frac{2y^{\frac{3}{2}}}{3} - \frac{y^3}{24} \right]_0^4$$

$$= \left(\frac{16}{3} - \frac{8}{3} \right)$$

$$= \frac{8}{3} \quad (\text{平方单位})$$

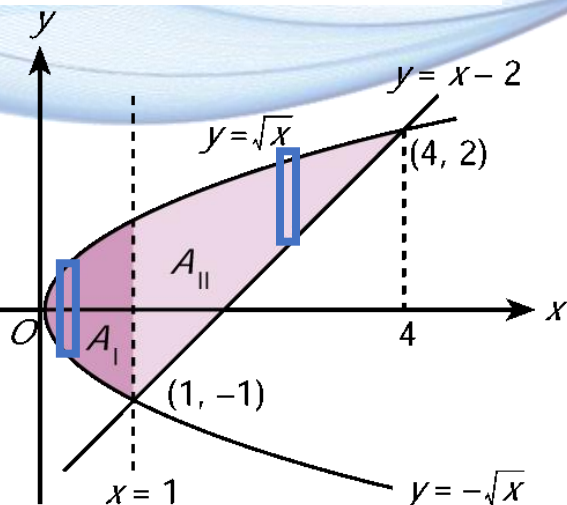
e)求曲线 $y^2 = x$ 与 $x = y + 2$ 所围成的面积。

解2:对x轴



解1:对y轴

$$\begin{aligned}
 A &= \int_{-1}^2 [(y+2) - y^2] dy \\
 &= \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_{-1}^2 \\
 &= \frac{10}{3} - \left(-\frac{7}{6} \right) \\
 &= \underline{\underline{\frac{9}{2}}} \text{ (单位平方)}
 \end{aligned}$$

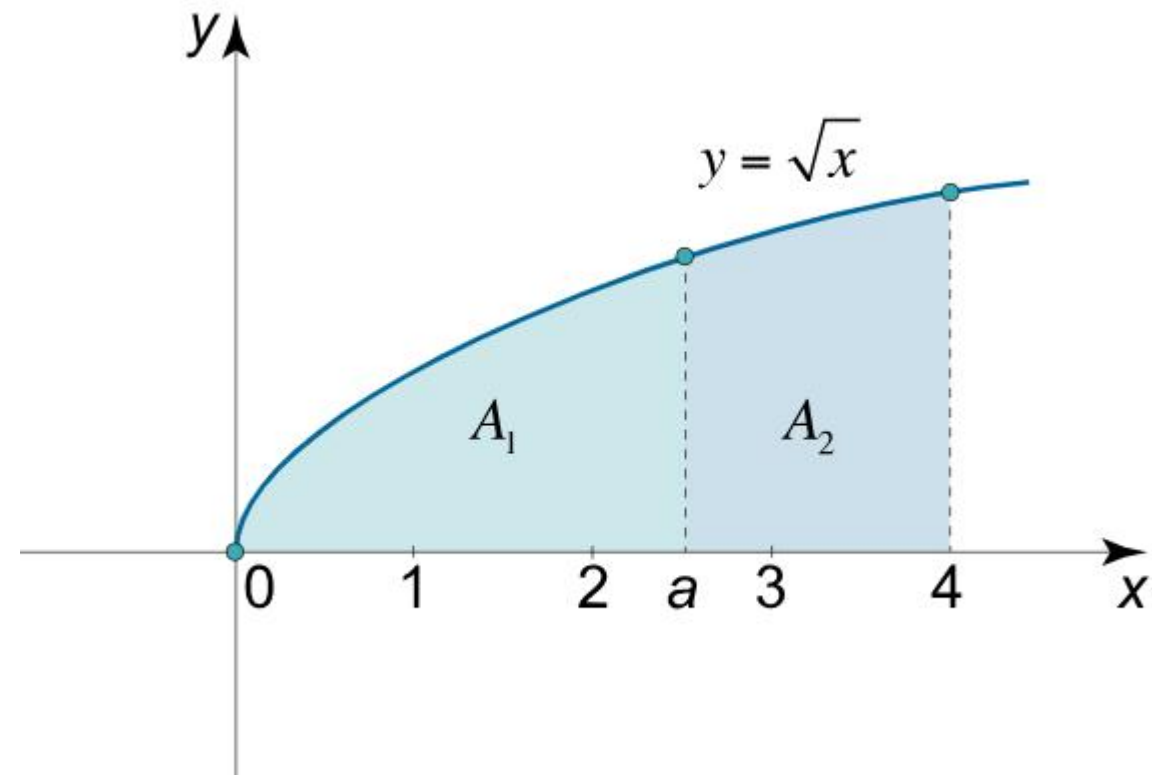


$$\begin{aligned}
 A_I &= \int_0^1 [\sqrt{x} - (-\sqrt{x})] dx \\
 &= 2 \int_0^1 x^{\frac{1}{2}} dx \\
 &= \left[\frac{4}{3} x^{\frac{3}{2}} \right]_0^1 \\
 &= \underline{\underline{\frac{4}{3}}}
 \end{aligned}$$

$$\begin{aligned}
 A_{II} &= \int_1^4 [\sqrt{x} - (x-2)] dx \\
 &= \left[\frac{2}{3} x^{\frac{3}{2}} - \frac{1}{2} x^2 + 2x \right]_1^4 \\
 &= \underline{\underline{\frac{19}{6}}}
 \end{aligned}$$

$$\begin{aligned}
 A &= A_I + A_{II} \\
 &= \frac{4}{3} + \frac{19}{6} \\
 &= \underline{\underline{\frac{9}{2}}} \text{ (单位平方)}
 \end{aligned}$$

f) Find the coordinate of the point a that splits the area under the root function $y = \sqrt{x}$ on the interval $[0, 4]$ into equal parts.



$$A_1 = A_2$$

$$\int_0^a \sqrt{x} dx = \int_a^4 \sqrt{x} dx$$

$$\left[\frac{2}{3} \sqrt{x^3} \right]_0^a = \left[\frac{2}{3} \sqrt{x^3} \right]_a^4$$

$$\frac{2}{3} \sqrt{a^3} - 0 = \frac{2}{3} \sqrt{4^3} - \frac{2}{3} \sqrt{a^3}$$

$$\frac{4}{3} \sqrt{a^3} = \frac{16}{3}$$

$$a^3 = 16$$

$$a = \sqrt[3]{16}$$

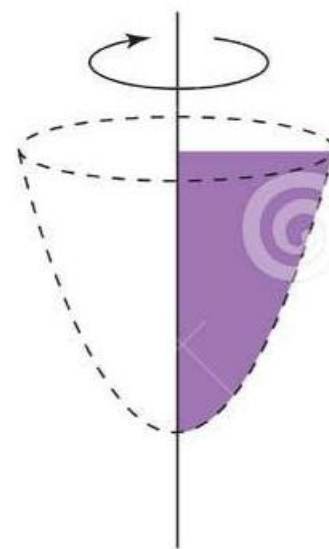
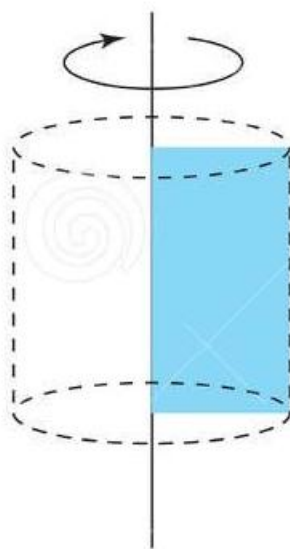
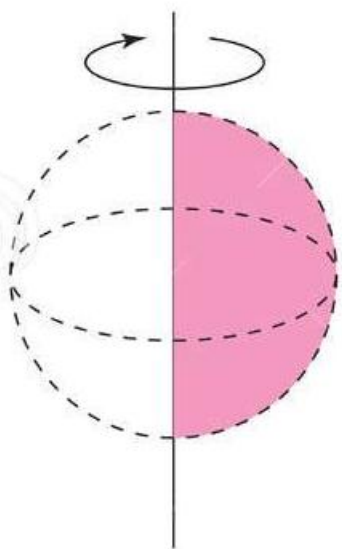
$$= 2\sqrt[3]{2}$$

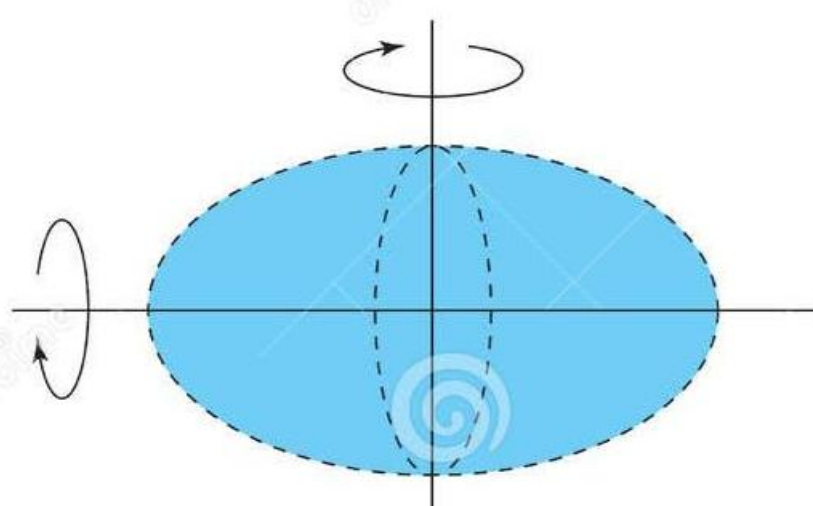
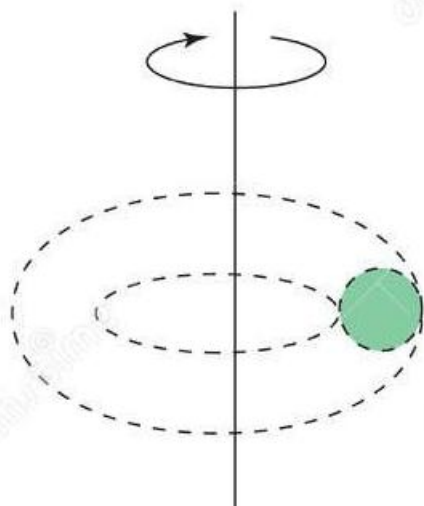
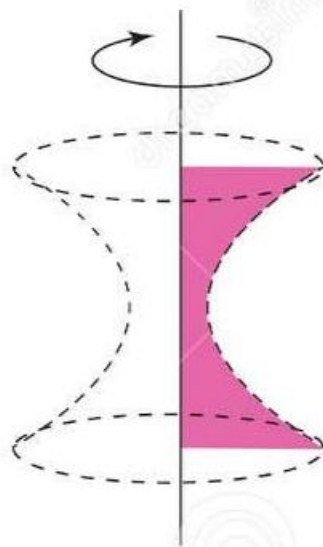
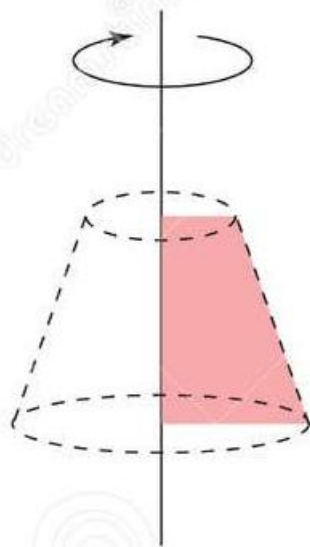
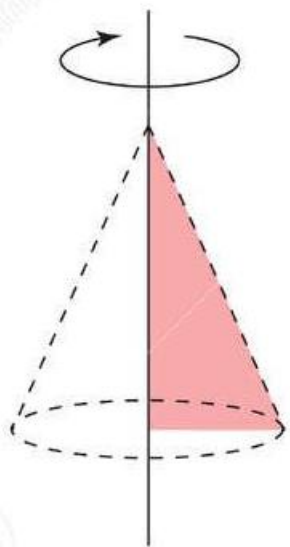
习题 24.3(2) - 9, 10, 11, 14

Chp24.4 旋转体体积

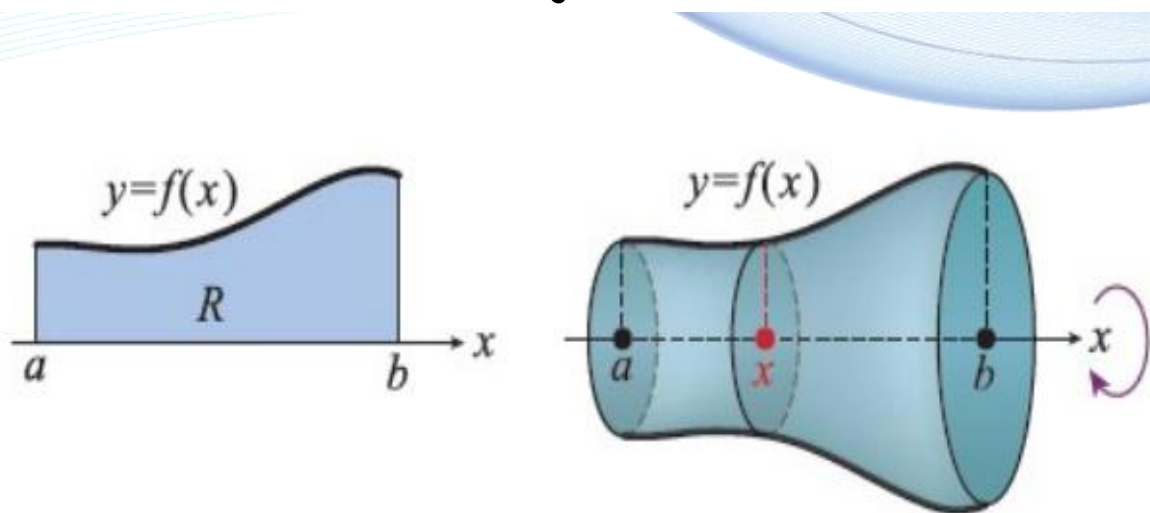
Pg. 142

旋转体就是由一个平面图形饶这平面内一条直线旋转一周而成的立体。这直线叫做**旋转轴**。



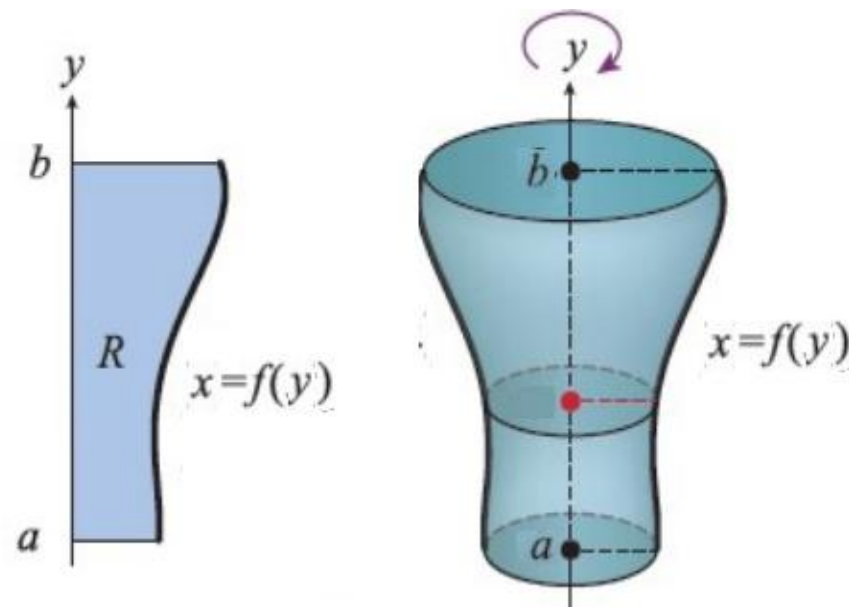


曲线绕 x, y 轴旋转所围成的旋转体体积公式



曲线 $y=f(x)$, $x=a$, $x=b$ 及 x 轴围成的区域绕 x 轴旋转的旋转体体积为

$$V = \int_a^b \pi [f(x)]^2 dx$$



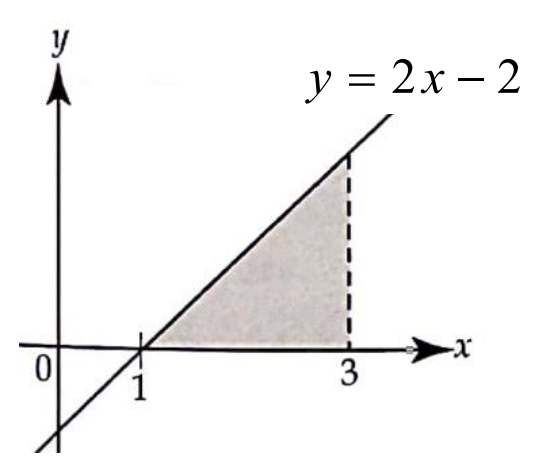
曲线 $x=f(y)$, $y=a$, $y=b$ 及 y 轴围成的区域绕 y 轴旋转的旋转体体积为

$$V = \int_a^b \pi [f(y)]^2 dy$$

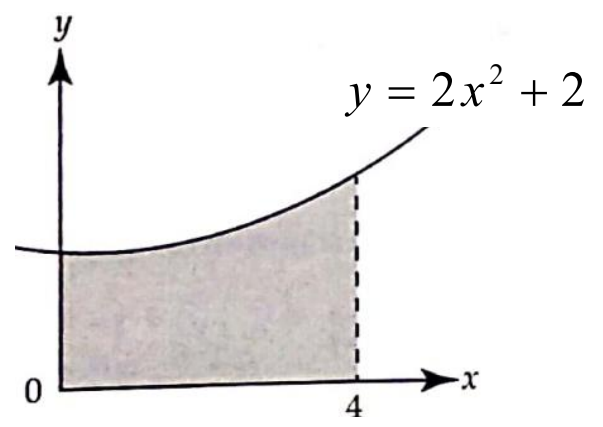
a) 用定积分表示下列图中各个阴影部分区域绕x轴旋转的

旋转体体积，并略绘出其旋转体体积图像：

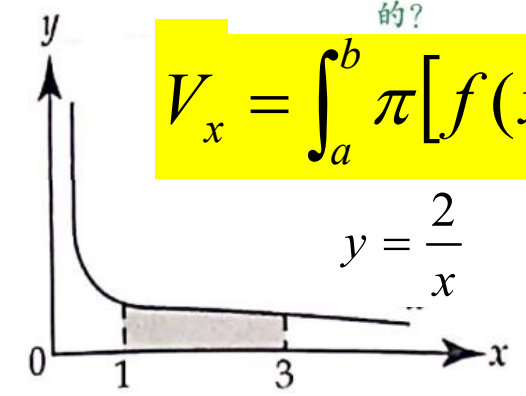
应用旋转体体积公式
 $V = \int_a^b \pi [f(x)]^2 dx$
 时， $f(x)$ 可否是负的？



$$V_x = \pi \int_1^3 (2x - 2)^2 dx$$



$$V_x = \pi \int_0^4 (2x^2 + 2)^2 dx$$

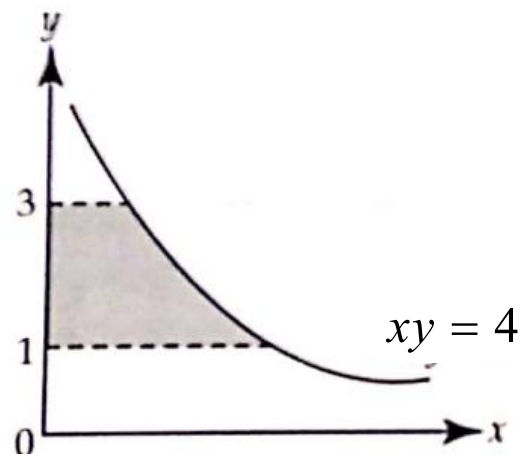


$$V_x = \pi \int_1^3 \left(\frac{2}{x}\right)^2 dx$$

$$V_x = \int_a^b \pi [f(x)]^2 dx$$

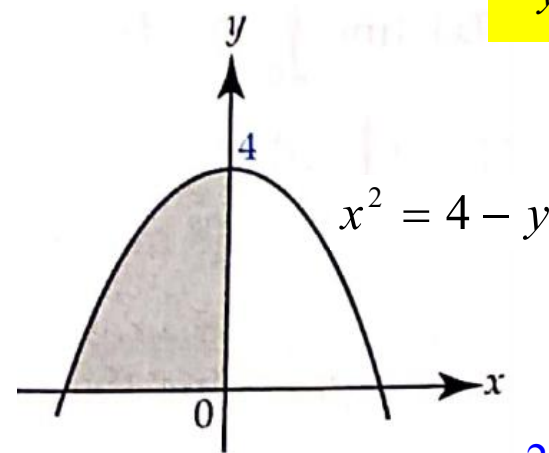
b) 用定积分表示下列图中各个阴影部分区域绕y轴旋转的旋转体体积，并略绘出其旋转体体积图像：

$$V_y = \int_a^b \pi [f(y)]^2 dy$$



$$xy = 4 \Rightarrow x = \frac{4}{y}$$

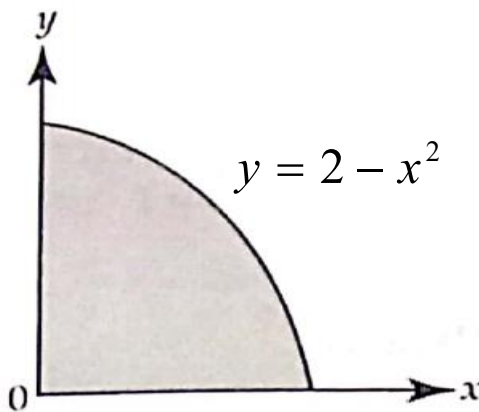
$$V_y = \pi \int_1^3 \left(\frac{4}{y} \right)^2 dy$$



$$x^2 = 4 - y$$

$$x = -\sqrt{4 - y} \text{ (取负)}$$

$$V_y = \pi \int_0^4 \left(-\sqrt{4 - y} \right)^2 dy$$



$$\text{令 } x=0 \text{ 代入 } y=2-x^2 \Rightarrow y=2$$

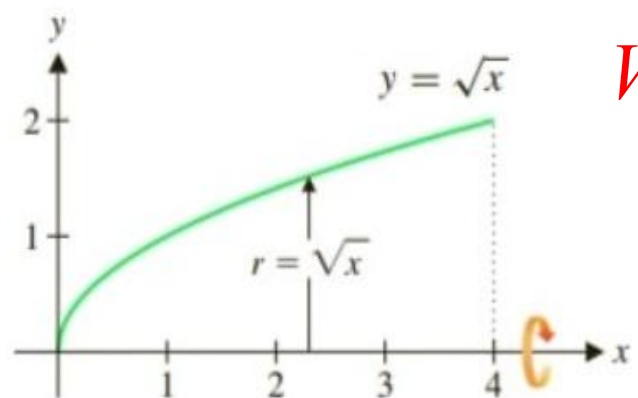
$$y = 2 - x^2$$

$$x = \sqrt{2 - y} \text{ (取正)}$$

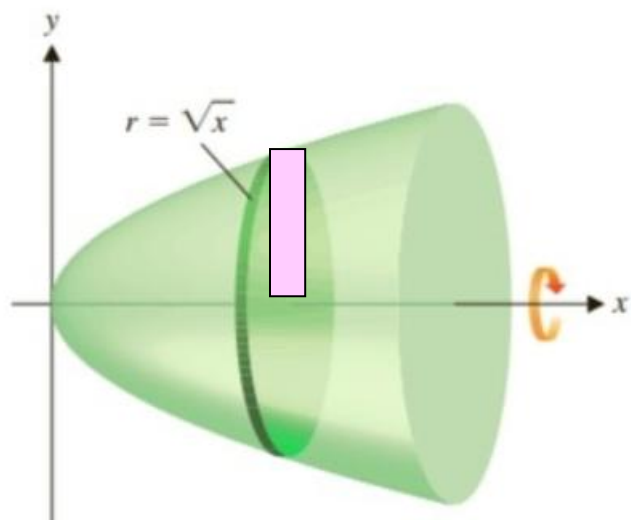
$$V_y = \pi \int_0^2 \left(\sqrt{2 - y} \right)^2 dy$$

c)如图，曲线 $y = \sqrt{x}$ 与 x 轴所围成的区域绕 x 轴旋转，其中 $0 \leq x \leq 4$ ，求所得轴旋体体积。

$$V = \int_a^b \pi [f(x)]^2 dx$$

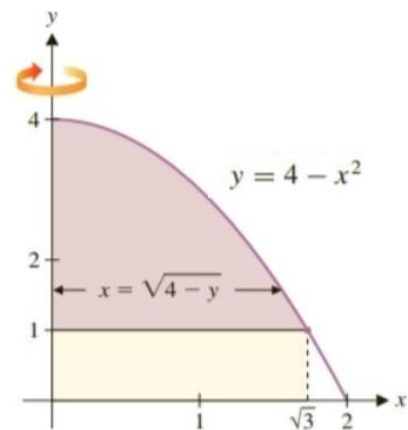


$$\begin{aligned} V_x &= \pi \int_0^4 (\sqrt{x})^2 dx \\ &= \pi \int_0^4 x dx \\ &= \pi \left[\frac{x^2}{2} \right]_0^4 \\ &= \pi [8 - 0] \\ &= 8\pi \quad (\text{立方单位}) \end{aligned}$$

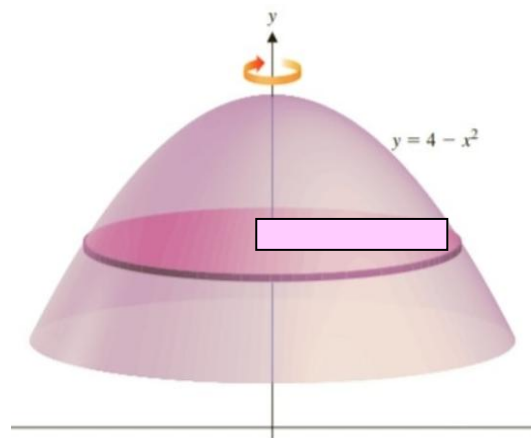


d)如图，曲线 $y = 4 - x^2$ 与 y 轴所围成的区域绕 y 轴旋转，其中 $1 \leq y \leq 4$ ，求所得旋转体体积。

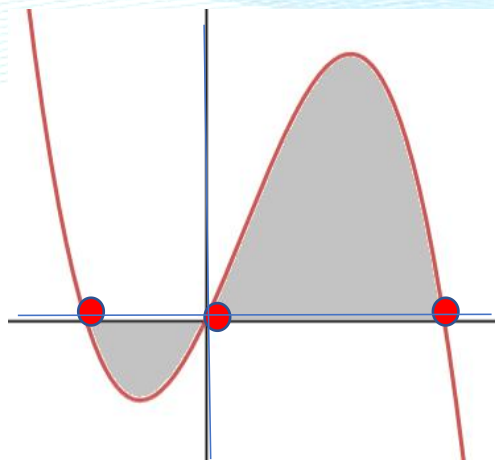
$$V = \int_a^b \pi [f(y)]^2 dy$$



$$\begin{aligned} y &= 4 - x^2 \Rightarrow x = \sqrt{4 - y} \\ V_y &= \pi \int_1^4 (\sqrt{4 - y})^2 dy \\ &= \pi \int_1^4 4 - y dy \\ &= \pi \left[4y - \frac{y^2}{2} \right]_1^4 \\ &= \pi \left[(16 - 8) - \left(4 - \frac{1}{2}\right) \right] \\ &= \frac{9}{2} \pi \quad (\text{立方单位}) \end{aligned}$$



e)求曲线 $y = -x^3 + x^2 + 2x$ 与 x 轴所围成的区域绕 x 轴旋转一周所得的旋转体体积。



$$\text{令 } -x(x-2)(x+1) = 0$$

$$x = 0 \quad \text{或} \quad x = 2 \quad \text{或} \quad x = -1$$

$$V_x = \pi \int_{-1}^0 (-x^3 + x^2 + 2x)^2 dx + \pi \int_0^2 (-x^3 + x^2 + 2x)^2 dx$$

$$= \pi \int_{-1}^2 (-x^3 + x^2 + 2x)^2 dx$$

$$= \pi \int_{-1}^2 (x^6 - 2x^5 - 3x^4 + 4x^3 + 4x^2) dx$$

$$= \pi \left[\frac{x^7}{7} - \frac{x^6}{3} - \frac{3x^5}{5} + x^4 + \frac{4x^3}{3} \right]_{-1}^2$$

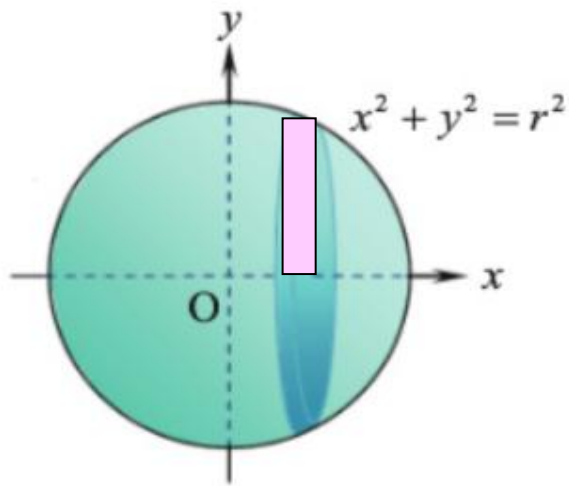
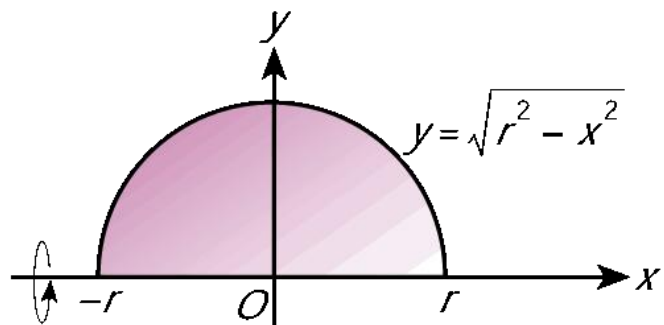
$$= \pi \left[\left(\frac{128}{7} - \frac{64}{3} - \frac{96}{5} + 16 + \frac{32}{3} \right) - \left(-\frac{1}{7} - \frac{1}{3} + \frac{3}{5} + 1 - \frac{4}{3} \right) \right]$$

$$= \frac{162}{35} \pi \quad (\text{立方单位})$$

习题24.4(1)-1,2,4,5

f)如图, 曲线 $y = \sqrt{r^2 - x^2}$ 与 x 轴所围成的区域绕 x 轴旋转, 其中 $r > 0, -r \leq x \leq r$, 证明所得的旋转体体积为 $\frac{4}{3}\pi r^3$ 。

$$V = \int_a^b \pi [f(x)]^2 dx$$



$$V_x = \pi \int_{-r}^r \left(\sqrt{r^2 - x^2} \right)^2 dx$$

$$= \pi \int_{-r}^r (r^2 - x^2) dx$$

$$= \pi \left[r^2 x - \frac{x^3}{3} \right]_{-r}^r$$

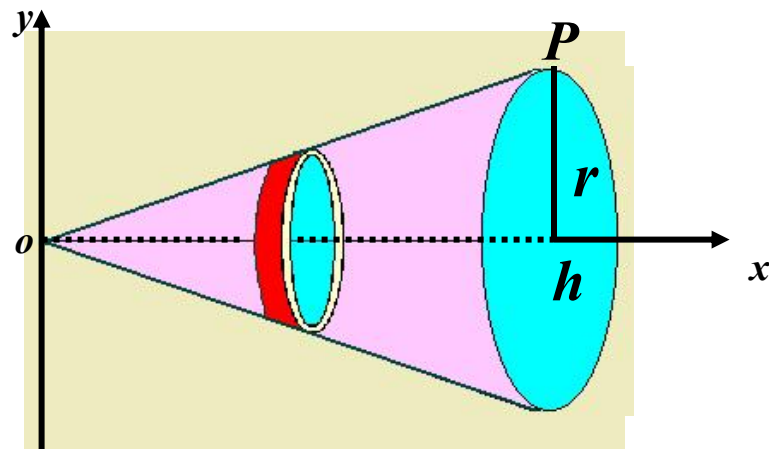
$$= \pi \left[\left(r^3 - \frac{r^3}{3} \right) - \left(-r^3 + \frac{r^3}{3} \right) \right]$$

$$= \frac{4}{3} \pi r^3 \quad (\text{立方单位}) \quad (\text{得证})$$

g) 直线 $y = \frac{r}{h}x$, $x = h$ 及 x 轴围成一个直角三角形, 绕 x 轴旋转构成一个底半径为 r ,

高为 h 的圆锥体, 证明圆锥体体积为 $\frac{1}{3}\pi r^2 h$ 。(其中 r, h 为常数)

$$V = \int_a^b \pi [f(x)]^2 dx$$



$$V_x = \pi \int_0^h \left(\frac{r}{h} x \right)^2 dx$$

$$= \frac{\pi r^2}{h^2} \int_0^h x^2 dx$$

$$= \frac{\pi r^2}{h^2} \left[\frac{x^3}{3} \right]_0^h$$

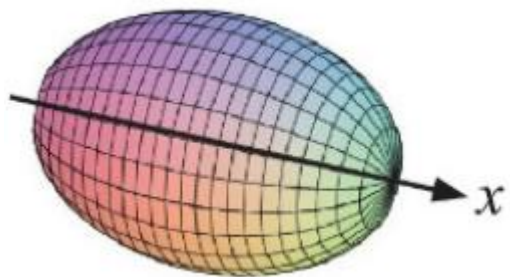
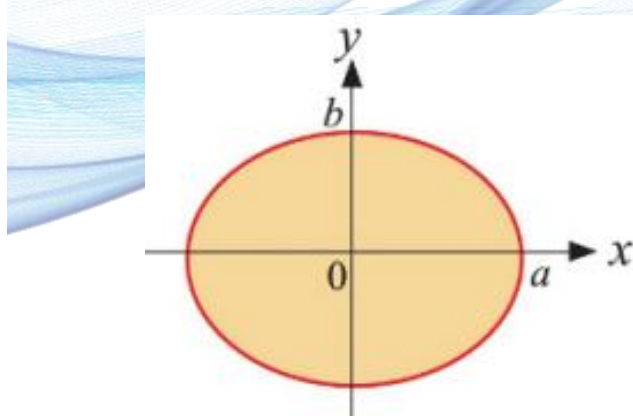
$$= \frac{\pi r^2}{h^2} \left(\frac{h^3}{3} \right)$$

$$= \frac{1}{3} \pi r^2 h \quad (\text{立方单位}) \quad (\text{得证})$$

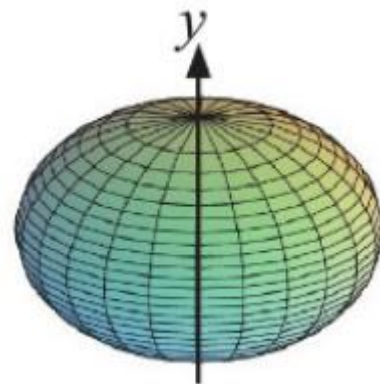
h) 右图为椭圆 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, 其中 $a, b > 0$, 求

i) 当椭圆绕 x 轴旋转一周, 证明椭圆球体体积为 $\frac{4}{3}\pi ab^2$ 。

ii) 当椭圆绕 y 轴旋转一周, 证明椭圆球体体积为 $\frac{4}{3}\pi a^2 b$ 。

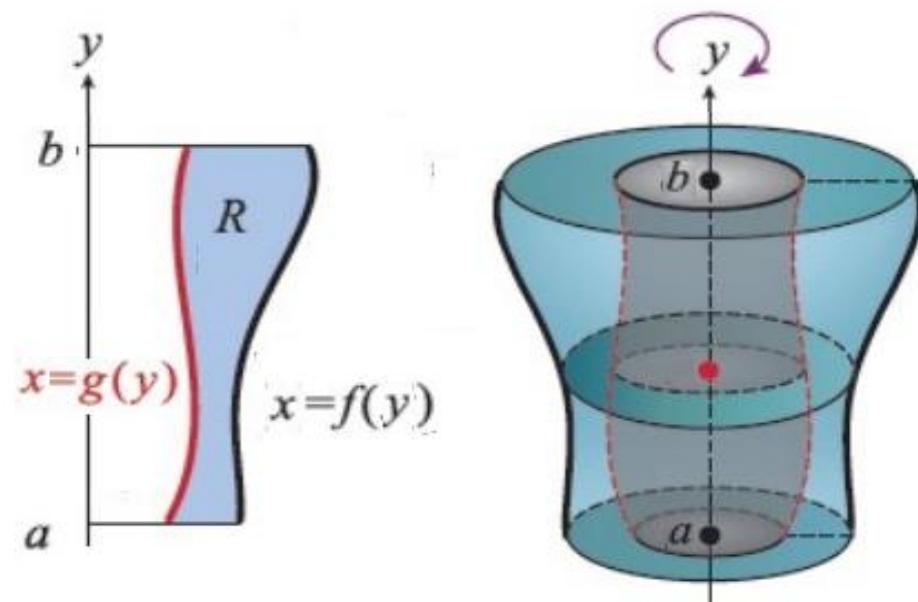
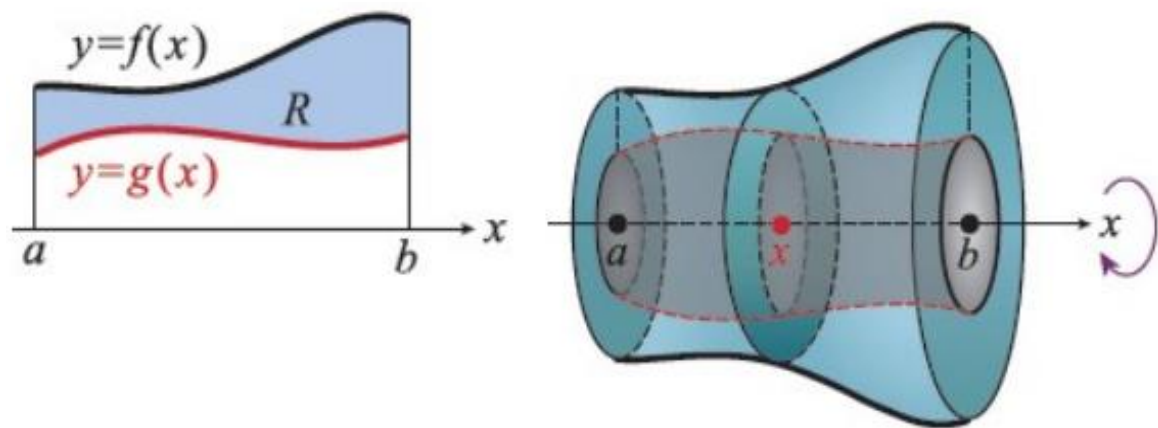


$$\begin{aligned} i) V_x &= \pi \int_{-a}^a b^2 \left(1 - \frac{x^2}{a^2} \right) dx \\ &= 2\pi \int_0^a b^2 \left(\frac{a^2 - x^2}{a^2} \right) dx \\ &= \frac{2b^2}{a^2} \pi \int_0^a (a^2 - x^2) dx \\ &= \frac{2b^2}{a^2} \pi \left[a^2 x - \frac{x^3}{3} \right]_0^a \\ &= \frac{2b^2}{a^2} \pi \left(a^3 - \frac{a^3}{3} \right) \\ &= \frac{4}{3} \pi ab^2 \quad (\text{立方单位}) \quad (\text{得证}) \end{aligned}$$



$$\begin{aligned} ii) V_y &= \pi \int_{-b}^b a^2 \left(1 - \frac{y^2}{b^2} \right) dy \\ &= 2\pi \int_0^b a^2 \left(\frac{b^2 - y^2}{b^2} \right) dy \\ &= \frac{2a^2}{b^2} \pi \int_0^b (b^2 - y^2) dy \\ &= \frac{2a^2}{b^2} \pi \left[b^2 y - \frac{y^3}{3} \right]_0^b \\ &= \frac{2a^2}{b^2} \pi \left(b^3 - \frac{b^3}{3} \right) \\ &= \frac{4}{3} \pi a^2 b \quad (\text{立方单位}) \quad (\text{得证}) \end{aligned}$$

两曲线所围成的区域绕 x, y 轴旋转一周所得的旋转体体积公式



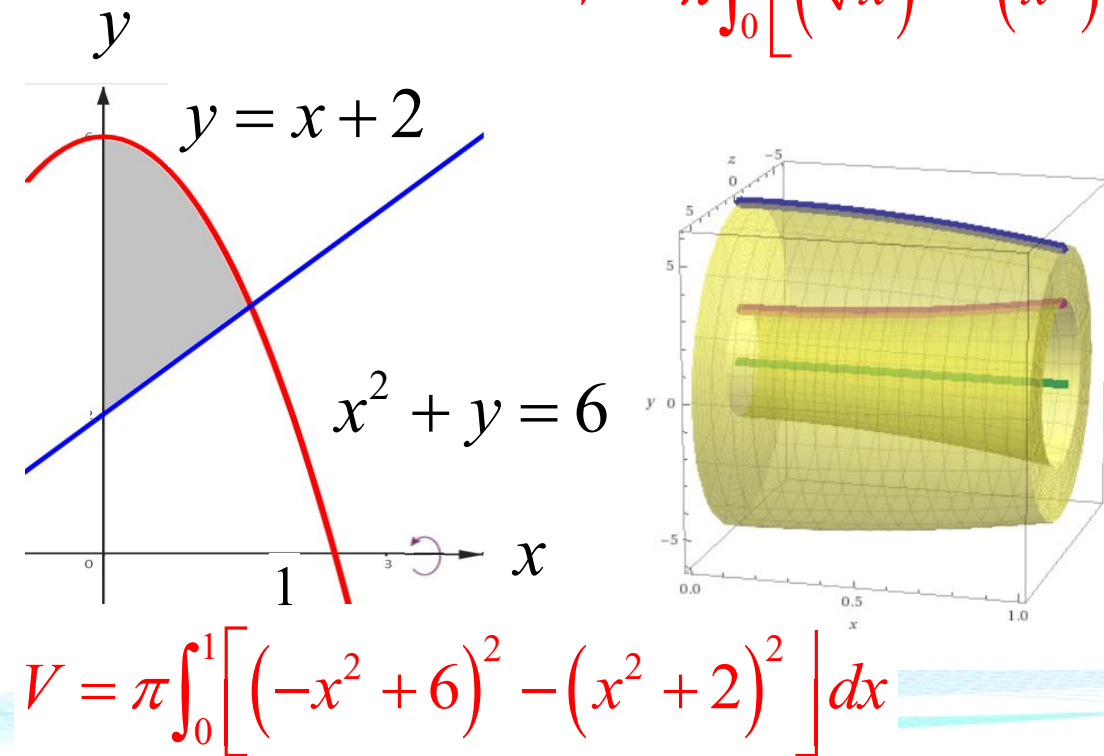
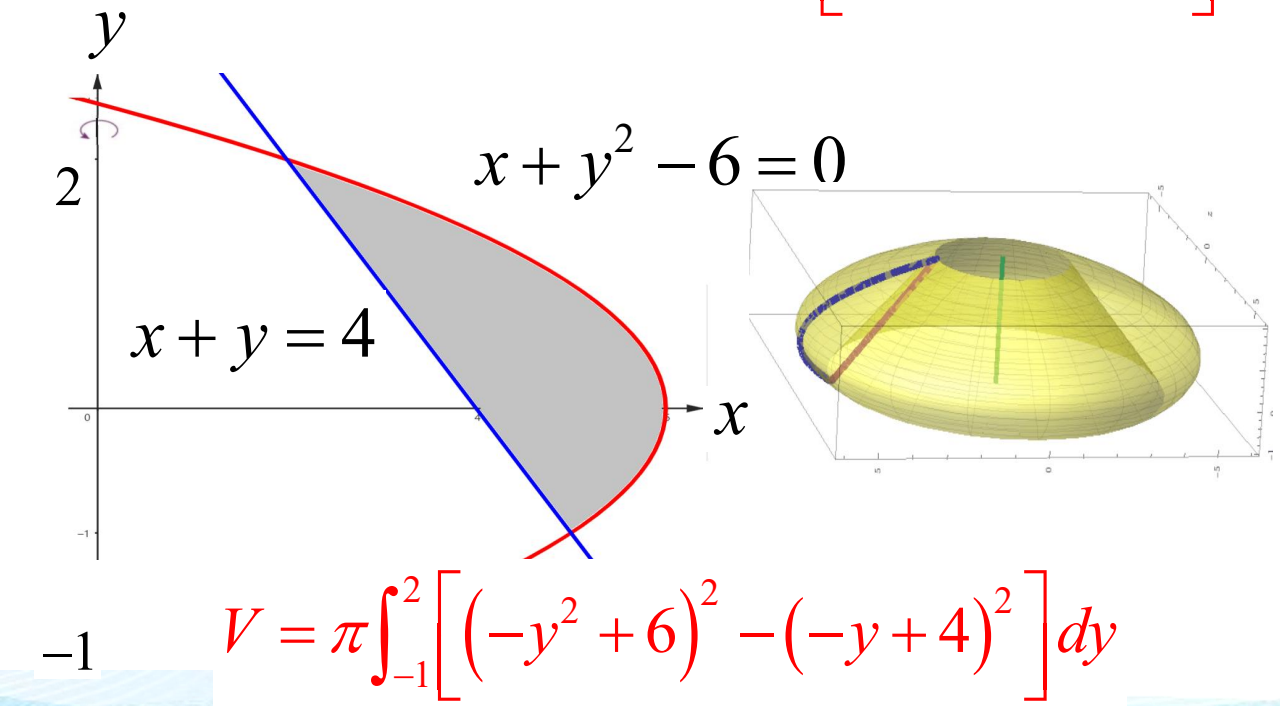
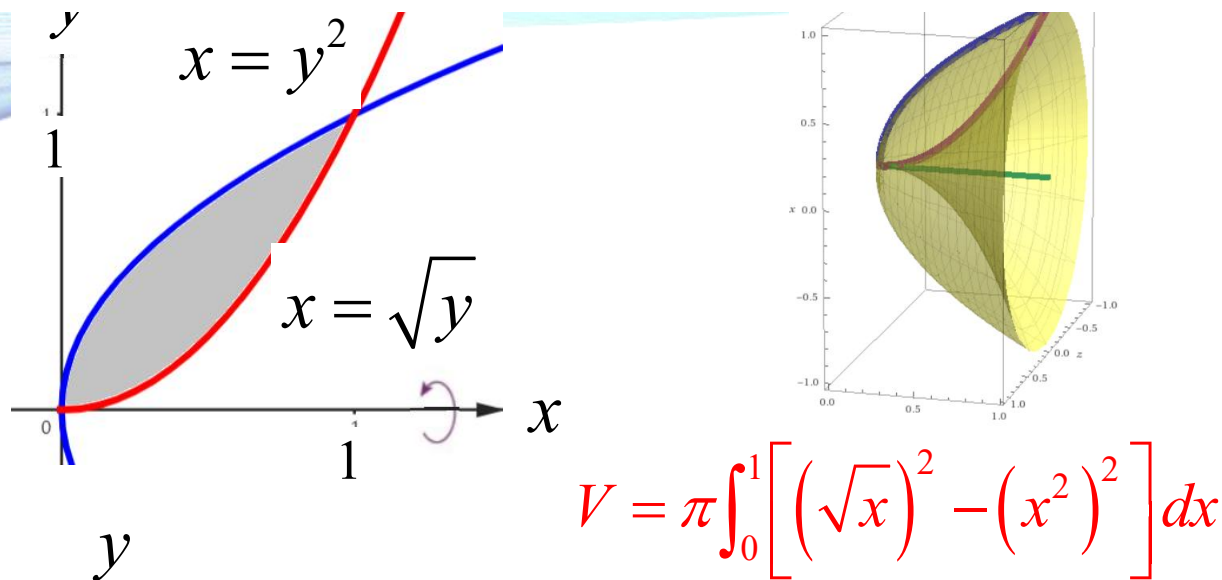
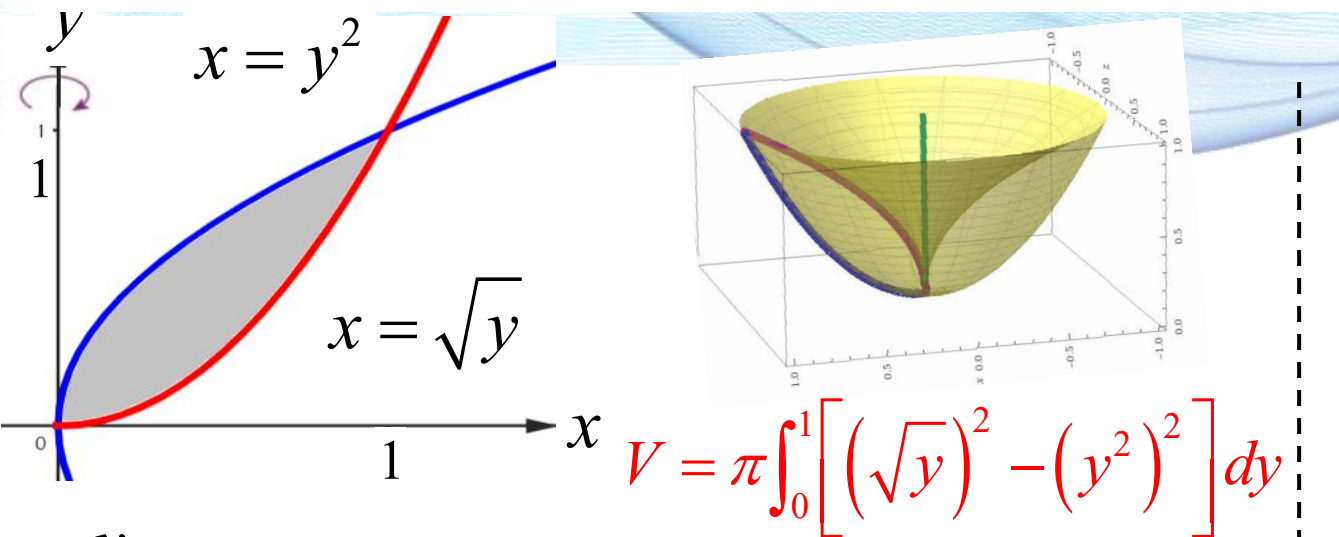
曲线 $y=f(x)$, $y=g(x)$, $x=a$, $x=b$ 及 x 轴围成的区域绕 x 轴旋转的旋转体体积为

$$V = \pi \int_a^b \left([f(x)]^2 - [g(x)]^2 \right) dx$$

曲线 $x=f(y)$, $x=g(y)$, $y=a$, $y=b$ 及 y 轴围成的区域绕 y 轴旋转的旋转体体积为

$$V = \pi \int_a^b \left([f(y)]^2 - [g(y)]^2 \right) dy$$

a) 用 定积分 表示下列图中各个阴影部分区域绕题目的要求旋转的旋转体体积:



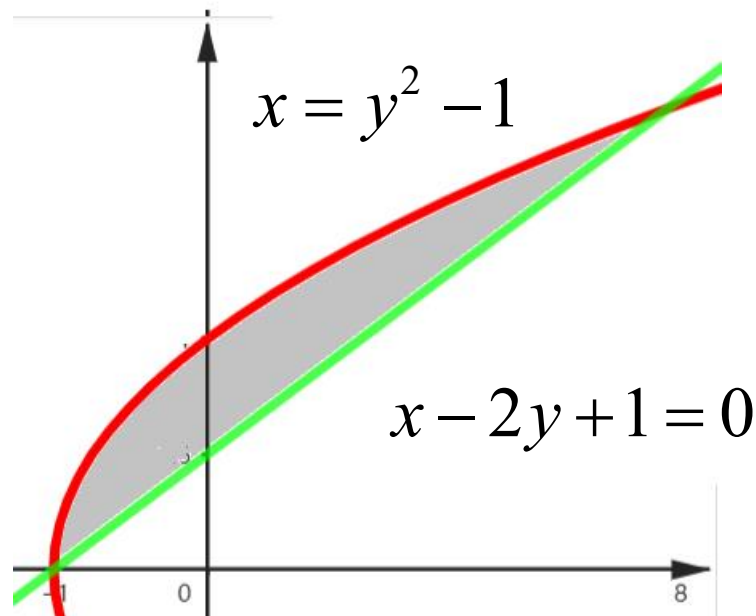
b)求 $x = y^2 - 1$ 与 $x - 2y + 1 = 0$ 所围成的面积绕 x 轴

旋转一周的旋转体体积。由 $x = y^2 - 1$

$$\Rightarrow y = \sqrt{x+1}$$

$$\text{由 } x - 2y + 1 = 0$$

$$\Rightarrow y = \frac{x+1}{2}$$



求交点就是要解联立方程

$$\begin{cases} x = y^2 - 1 & \text{---(1) 绕 } x \text{ 轴旋转,} \\ x - 2y + 1 = 0 & \text{---(2) 仅需要 } x \text{ 的上下限} \end{cases}$$

解得 $x = -1, x = 3$

$$V = \pi \int_{-1}^3 \left[\left(\sqrt{x+1} \right)^2 - \left(\frac{x+1}{2} \right)^2 \right] dx$$

$$= \pi \int_{-1}^3 \left[x+1 - \frac{x^2 + 2x + 1}{4} \right] dx$$

$$= \pi \int_{-1}^3 \left[-\frac{x^2}{4} + \frac{x}{2} + \frac{3}{4} \right] dx$$

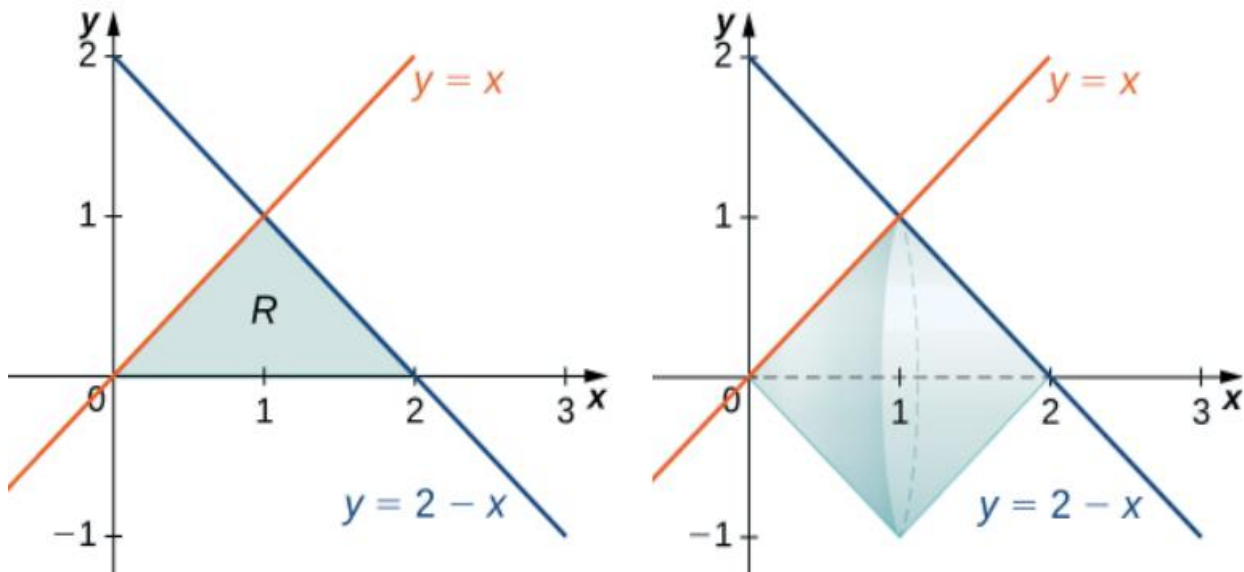
$$= \pi \left[-\frac{x^3}{12} + \frac{x^2}{4} + \frac{3x}{4} \right]_{-1}^3$$

$$= \pi \left(-\frac{27}{12} + \frac{9}{4} + \frac{9}{4} - \left(-\frac{1}{12} + \frac{1}{4} - \frac{3}{4} \right) \right)$$

$$= \frac{8\pi}{3} \text{ (立方单位)}$$

c)求 $y = x$, $y = 2 - x$ 与 x 轴所围成的面积绕 x 轴

旋转一周的旋转体体积。



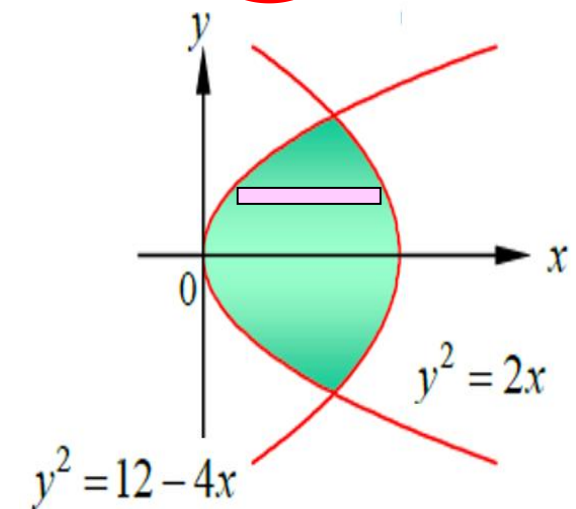
$$\begin{cases} y = x & \text{---(1)} \\ y = 2 - x & \text{---(2)} \end{cases}$$

解得 $x = 1$

$$\begin{aligned} V &= \pi \int_0^1 (x)^2 dx + \pi \int_1^2 (2-x)^2 dx \\ &= \pi \left[\frac{x^3}{3} \right]_0^1 + \pi \int_1^2 (4 - 4x + x^2) dx \\ &= \pi \left(\frac{1}{3} \right) + \pi \left[4x - 2x^2 + \frac{x^3}{3} \right]_1^2 \\ &= \frac{\pi}{3} + \pi \left(\left(8 - 8 + \frac{8}{3} \right) - \left(4 - 2 + \frac{1}{3} \right) \right) \\ &= \frac{2\pi}{3} \text{ (立方单位)} \end{aligned}$$

d)如图, 求曲线 $y^2 = 2x$ 与 $y^2 = 12 - 4x$ 所围成的区域

绕 y 轴旋转一周所得的旋转体体积。



$$\begin{cases} y^2 = 2x & \text{---(1)} \\ y^2 = 12 - 4x & \text{---(2)} \end{cases}$$

由(1), $x = \frac{y^2}{2}$ 代入(2),

$$y^2 = 12 - 4\left(\frac{y^2}{2}\right)$$

$$y^2 = 4$$

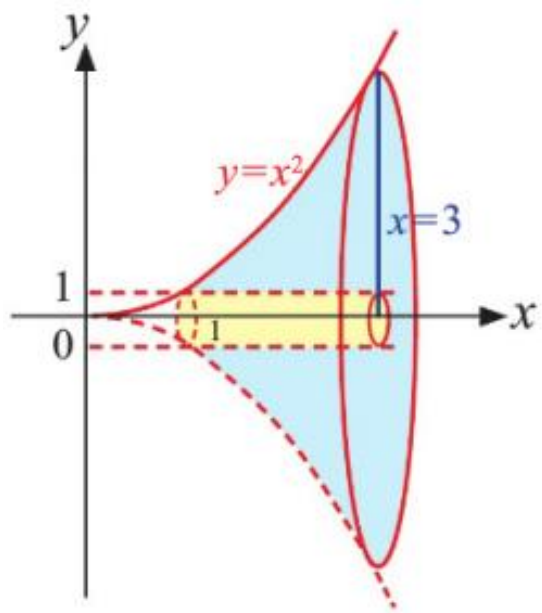
$$y = \pm 2$$

$$y^2 = 2x \Rightarrow x = \frac{y^2}{2}$$

$$y^2 = 12 - 4x \Rightarrow x = 3 - \frac{y^2}{4}$$

$$\begin{aligned} V_y &= \pi \int_{-2}^2 \left[\left(3 - \frac{y^2}{4} \right)^2 - \left(\frac{y^2}{2} \right)^2 \right] dy \\ &= \pi \int_{-2}^2 \left(9 - \frac{3y^2}{2} - \frac{3y^4}{16} \right) dy \\ &= \pi \left[9y - \frac{y^3}{2} - \frac{3y^5}{80} \right]_{-2}^2 \\ &= \pi \left[\left(18 - 4 - \frac{6}{5} \right) - \left(-18 + 4 + \frac{6}{5} \right) \right] \\ &= \frac{128}{5} \pi \quad (\text{立方单位}) \end{aligned}$$

e)求曲线 $y = x^2$ 直线 $x = 3$ 及 $y = 1$ 所围成的区域
绕 x 轴旋转一周所得的旋转体体积。



$$\begin{cases} y = x^2 & \text{---(1)} \\ y = 1 & \text{---(2)} \end{cases}$$

$$(2) \text{ 代入 } (1), x^2 = 1$$

$$x = 1 \quad \text{或} \quad x = -1$$

观看圆绕 y 轴旋
转一周形成环面



<https://youtu.be/VlaQhKOAf3I>

环面的另一种形
成方式

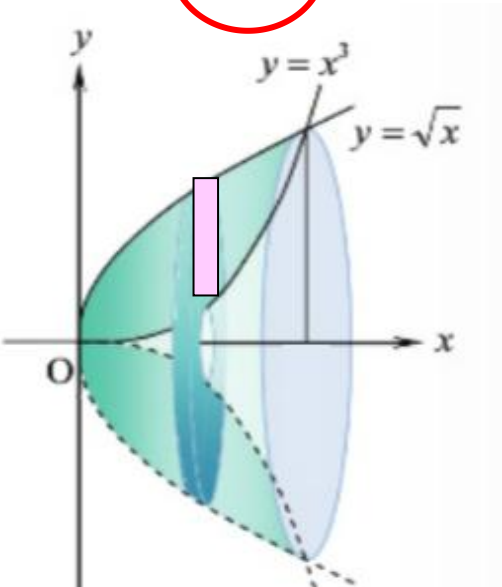


<https://youtu.be/qzkuNpJJfIU>

$$\begin{aligned} V_x &= \pi \int_1^3 \left[(x^2)^2 - 1 \right] dx \\ &= \pi \int_1^3 (x^4 - 1) dx \\ &= \pi \left[\frac{x^5}{5} - x \right]_1^3 \\ &= \pi \left[\left(\frac{243}{5} - 3 \right) - \left(\frac{1}{5} - 1 \right) \right] \\ &= \frac{232}{5} \pi \quad (\text{立方单位}) \end{aligned}$$

f) 已知一区域由曲线 $y = x^3$ 与 $y = \sqrt{x}$ 所围成。求此区域

i) 绕 x 轴旋转一周所得的旋转体体积;



$$\begin{cases} y = x^3 & \text{--- (1)} \\ y = \sqrt{x} & \text{--- (2)} \end{cases}$$

(2) 代入 (1), $x^3 = \sqrt{x}$

$$x^6 - x = 0$$

$$x(x^5 - 1) = 0$$

$$x = 0 \quad \text{或} \quad x = 1$$

$$V_x = \pi \int_0^1 \left((\sqrt{x})^2 - (x^3)^2 \right) dx$$

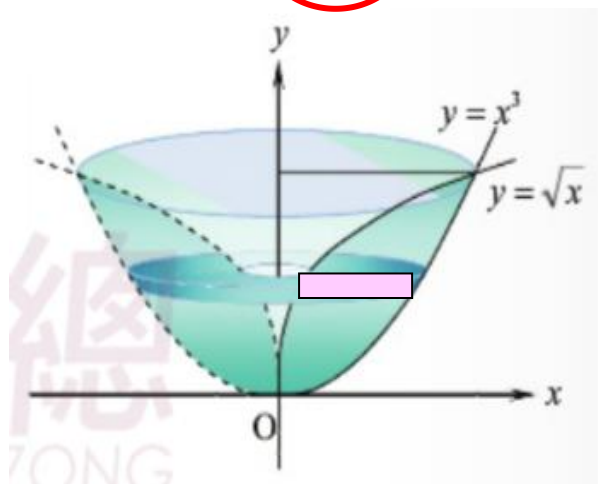
$$= \pi \int_0^1 (x - x^6) dx$$

$$= \pi \left[\frac{x^2}{2} - \frac{x^7}{7} \right]_0^1$$

$$= \pi \left[\left(\frac{1}{2} - \frac{1}{7} \right) - 0 \right]$$

$$= \frac{5}{14} \pi \quad (\text{立方单位})$$

ii) 绕 y 轴旋转一周所得的旋转体体积。



已知 $x = 0$ 或 $x = 1$ 代入 (1),
得 $y = 0$ 或 $y = 1$

$$y = x^3 \Rightarrow x = y^{\frac{1}{3}}$$

$$y = \sqrt{x} \Rightarrow x = y^2$$

$$V_y = \pi \int_0^1 \left(\left(y^{\frac{1}{3}} \right)^2 - (y^2)^2 \right) dy$$

$$= \pi \int_0^1 \left(y^{\frac{2}{3}} - y^4 \right) dy$$

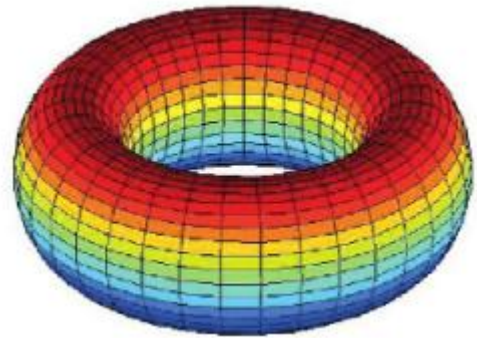
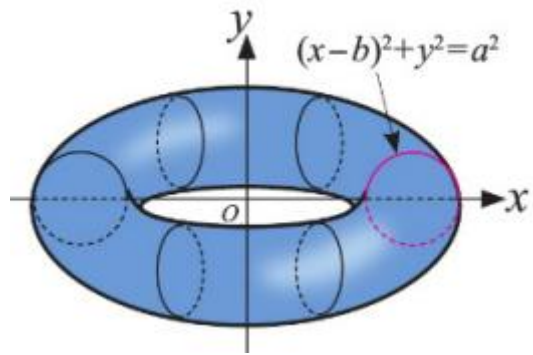
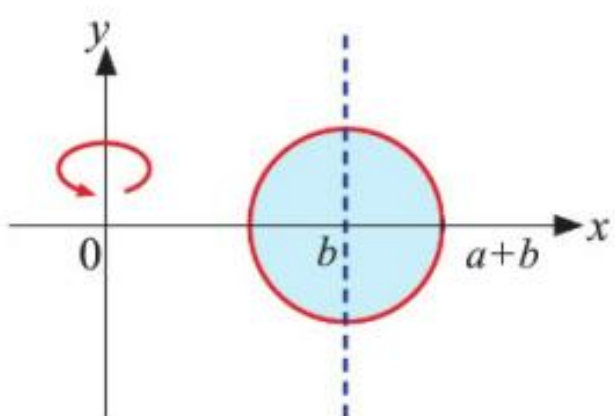
$$= \pi \left[\frac{y^{\frac{5}{3}}}{\frac{5}{3}} - \frac{y^5}{5} \right]_0^1$$

$$= \pi \left[\left(\frac{3}{5} - \frac{1}{5} \right) - 0 \right]$$

$$= \frac{2}{5} \pi \quad (\text{立方单位})$$

习题 24.4(2) - 6, 7, 10

g) 求图中圆 $(x-b)^2 + y^2 = a^2$ 绕 y 轴旋转一周的旋转体体积, 其中 $b > a$ 。



$$(x-b)^2 + y^2 = a^2$$

$$x-b = \pm\sqrt{a^2 - y^2}$$

$$x = b \pm \sqrt{a^2 - y^2} \quad \text{即 } x_{\text{右}} = b + \sqrt{a^2 - y^2}, \quad x_{\text{左}} = b - \sqrt{a^2 - y^2}$$

$$V_y = \pi \int_{-a}^a \left[\left(b + \sqrt{a^2 - y^2} \right)^2 - \left(b - \sqrt{a^2 - y^2} \right)^2 \right] dy$$

$$= \pi \int_{-a}^a \left[\left(b + \sqrt{a^2 - y^2} \right) + \left(b - \sqrt{a^2 - y^2} \right) \right] \left[\left(b + \sqrt{a^2 - y^2} \right) - \left(b - \sqrt{a^2 - y^2} \right) \right] dy$$

$$= \pi \int_{-a}^a 2b \left(2\sqrt{a^2 - y^2} \right) dy$$

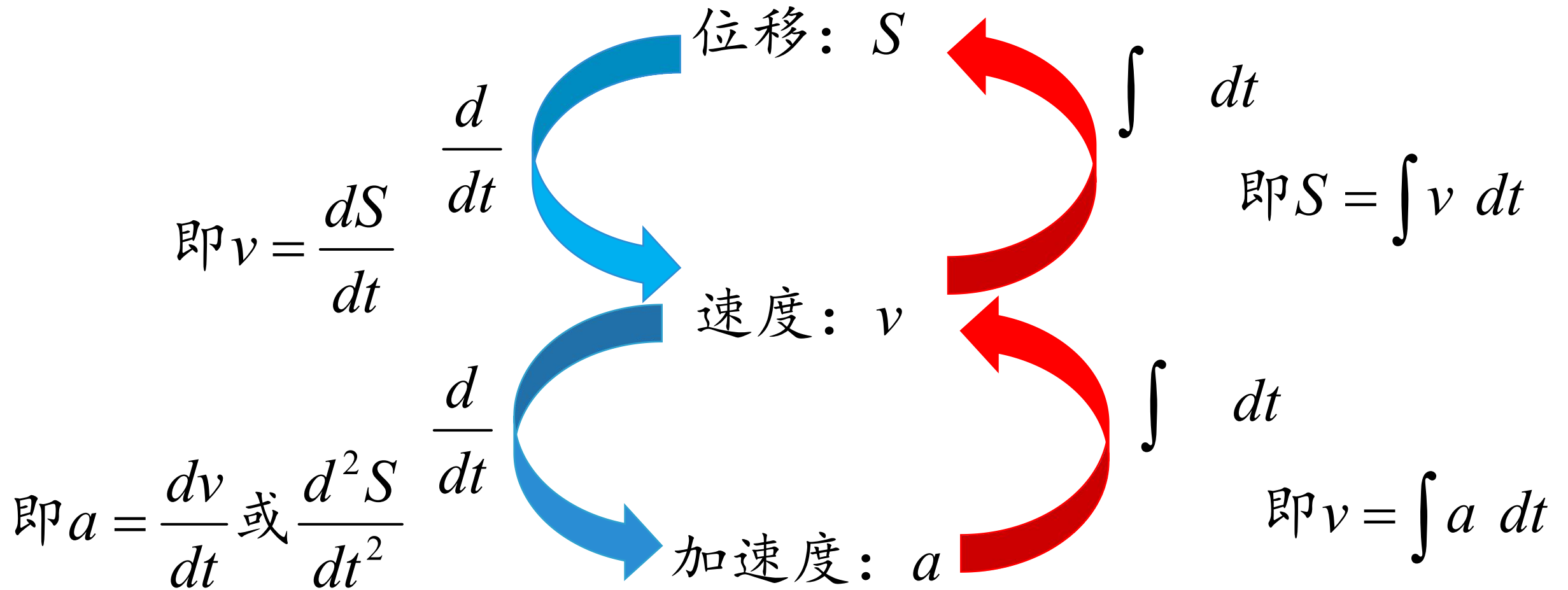
$$= 4b\pi \int_{-a}^a \sqrt{a^2 - y^2} dy$$

$$= 4b\pi \left(\frac{\pi a^2}{2} \right) \quad \because \int_{-a}^a \sqrt{a^2 - y^2} dy = \frac{\pi a^2}{2} \text{ 为半径为 } a \text{ 的半圆面积}$$

$$= 2\pi^2 a^2 b \quad (\text{立方单位})$$

Chp24.5 直线运动问题

Pg. 154



即 $t=0$ 时, $v=0$

a) 一物体由静止开始运动, 在 t 秒末, 其速度为 $5t^2 \text{ ms}^{-1}$; 问3秒后, 它离出发点有多远? 它费了多少时间才走了 360m ?

解1: 不定积分

$$S(t) = \int (5t^2) dt$$

$$= \frac{5t^3}{3} + C$$

当 $t=0$,
在质点出发,
 $S=0$

$$\because t=0, S=0$$

$$\therefore C=0$$

$$S(t) = \frac{5t^3}{3}$$

$$\begin{aligned} \text{3秒后, } S &= S(3) - S(0) \\ &= 45\text{m} \end{aligned}$$

$$360 = S(t_1) - S(0)$$

$$360 = \frac{5t_1^3}{3} - 0$$

$$t_1^3 = 216$$

$$t = 6\text{s}$$

解2: 定积分

$$S = S(3) - S(0)$$

$$= \int_0^3 (5t^2) dt$$

$$= \left[\frac{5t^3}{3} \right]_0^3$$

$$= 45\text{m}$$

$$360 = S(t_1) - S(0)$$

$$360 = \int_0^{t_1} v dt$$

$$360 = \int_0^{t_1} (5t^2) dt$$

$$360 = \left[\frac{5t^3}{3} \right]_0^{t_1}$$

$$360 = \frac{5t_1^3}{3}$$

$$t_1^3 = 216$$

$$t = 6\text{s}$$

b)一物体作直线运动, 其速度为 $v=t(4-t^2)ms^{-1}$ 。

求在前3秒内此物体所经过的总路程及其平均速率。解2:定积分

解1:不定积分 $S(t)=\int t(4-t^2)dt$

令 $v=0$

$$t(4-t^2)=0$$

$t=0$ 或 $t=2$ 或 $t=-2$ (不合)

当 $t=0$,
在质点出发,
 $S=0$

$$=\int (4t-t^3)dt$$

$$=2t^2-\frac{t^4}{4}+C$$

$$\because t=0, S=0$$

$$\therefore C=0$$

$$S(t)=2t^2-\frac{t^4}{4}$$

$$\text{总路程}=|S(2)-S(0)|+|S(3)-S(2)|$$

$$=|(8-4)-0|+\left|\left(18-\frac{81}{4}\right)-(8-4)\right|$$

$$=4+\frac{25}{4}$$

$$=\frac{41}{4}m$$

$$\text{总路程}=|S(2)-S(0)|+|S(3)-S(2)|$$

$$=\left|\int_0^2 (4t-t^3)dt\right|+\left|\int_2^3 (4t-t^3)dt\right|$$

$$=\left[2t^2-\frac{t^4}{4}\right]_0^2+\left[2t^2-\frac{t^4}{4}\right]_2^3$$

$$=|(8-4)-0|+\left|\left(18-\frac{81}{4}\right)-(8-4)\right|$$

$$=4+\frac{25}{4}$$

$$=\frac{41}{4}m$$

$$\text{平均速率}=\frac{41}{4} \div 3 = \frac{41}{12}ms^{-1}$$

c)一质点作直线运动，其通过点 O 的 t 秒后速度为 $v = (3t^2 - 10t + 3) \text{ ms}^{-1}$ 。

若此质点第一次瞬间静止于点 A ，求

a)质点到达点 A 的时间；

b) OA 的距离；

$$a)v = 3t^2 - 10t + 3$$

$$\text{令 } v = 0$$

$$3t^2 - 10t + 3 = 0$$

$$(3t - 1)(t - 3) = 0$$

$$t = \frac{1}{3} \text{ 或 } t = 3 (\text{不合})$$

$$\therefore \text{质点到达 } A \text{ 点的时间为 } \frac{1}{3} \text{ s}$$

解1: 定积分

$$b) S = S\left(\frac{1}{3}\right) - S(0)$$

$$= \int_0^{\frac{1}{3}} v dt$$

$$= \int_0^{\frac{1}{3}} (3t^2 - 10t + 3) dt$$

$$= \left[t^3 - 5t^2 + 3t \right]_0^{\frac{1}{3}}$$

$$= \left[\left(\frac{1}{3}\right)^3 - 5\left(\frac{1}{3}\right)^2 + 3\left(\frac{1}{3}\right) \right] - 0$$

$$= \frac{13}{27} \text{ m}$$

解2: 不定积分

⋮

c)一质点作直线运动, 其通过点 O 的 t 秒后速度为 $v=(3t^2-10t+3)ms^{-1}$ 。

若此质点第一次瞬间静止于点 A , 求

c)当加速度为零时, 此质点的速度; d)此质点于第二次瞬间停止时所走过的总路程。

$$c)v=3t^2-10t+3$$

$$a=6t-10$$

$$\text{令 } a=0$$

$$6t-10=0$$

$$t=\frac{5}{3} \text{ 代入}$$

$$v=3\left(\frac{5}{3}\right)^2-10\left(\frac{5}{3}\right)+3$$

$$v=-\frac{16}{3}m/s$$

解1: 定积分

$$d)\text{总路程}=\left|S\left(\frac{1}{3}\right)-S(0)\right|+\left|S(3)-S\left(\frac{1}{3}\right)\right|$$

$$=\frac{13}{27}+\left|\int_{\frac{1}{3}}^3 v dt\right|$$

$$=\frac{13}{27}+\left|\int_{\frac{1}{3}}^3 (3t^2-10t+3)dt\right|$$

$$=\frac{13}{27}+\left|t^3-5t^2+3t\right|_{\frac{1}{3}}^3$$

$$=\frac{13}{27}+\left|(27-45+9)-\left(\left(\frac{1}{3}\right)^3-5\left(\frac{1}{3}\right)^2+3\left(\frac{1}{3}\right)\right)\right|$$

$$=\frac{13}{27}+\frac{256}{27}$$

$$=\frac{269}{27}m$$

解2: 不定积分

⋮

习题 24.5-4,6,9

微积分基本定理

a) $f(x) = \int_2^x (3t+4) dt$, 求 $f'(2)$ 的值。

$$f(x) = \int_2^x (3t+4) dt$$

$$= \left[\frac{3t^2}{2} + 4t \right]_2^x$$

$$= \frac{3x^2}{2} + 4x - 14$$

$$f'(x) = 3x + 4$$

$$f'(2) = 10$$

结论

$$f(x) = \int_a^x f(t) dt$$

$$\text{则 } f'(x) = f(x)$$

$$g(x) = \int_a^x f(t) dt \Rightarrow g'(x) = f(x), \text{ 其中 } a \leq x \leq b$$

b) $f(x) = \int_0^x (t^3 - 5t + 6) dt$, 则 $f'(x) = \underline{x^3 - 5x + 6}$ 。

c) $f(x) = \int_0^x t^2 \sin\left(\frac{\pi}{2}t\right) dt$, 则 $f'(x) = \underline{x^2 \sin\left(\frac{\pi}{2}x\right)}$ 。

d) $F(x) = \int_5^x \frac{\sqrt{t^2+9}}{t} dt$, 则 $F'(x) = \underline{\frac{\sqrt{x^2+9}}{x}}$;
 $F'(\pi-1) = \underline{\frac{\sqrt{(\pi-1)^2+9}}{\pi-1}}$ 。

e) $f(x) = \int_x^{10} \frac{1}{\sqrt{2+t^3}} dt$, 则 $f'(x) = \underline{-\frac{1}{\sqrt{2+x^3}}}$ 。

f)已知 $\int_c^x f(t)dt = x^2 - 2x - 3$, 求 $f(x)$ 及常数 c 。

$$\because \left[\int_c^x f(t)dt \right]' = f(x)$$

$$\therefore [x^2 - 2x - 3]' = f(x)$$

$$f(x) = 2x - 2$$

$$\text{由} \int_c^x f(t)dt = x^2 - 2x - 3$$

当 $x=c$ 代入上式,

$$\int_c^c f(t)dt = c^2 - 2c - 3$$

$$0 = c^2 - 2c - 3$$

$$(c-3)(c+1) = 0$$

$$c = 3 \quad \text{或} \quad c = -1$$

$$\because \int_c^c f(t)dt = 0$$

练)已知 $\int_a^x f(t)dt = x^2 - 3x - 10$, 求 $f(x)$ 及常数 a 。 【 $2x - 3, a = 5, -2$ 】

g) 已知 $xf(x) = \frac{3}{2}x^4 - 4x^2 + 6 + \int_2^x f(t)dt$, 求 $f(x)$ 。

$$xf(x) = \frac{3}{2}x^4 - 4x^2 + 6 + \int_2^x f(t)dt \quad \text{当 } x=2 \text{ 代入}$$

二边对 x 求导,

$$\cancel{f(x)} + xf'(x) = 6x^3 - 8x + \cancel{f(x)}$$

$$f'(x) = 6x^2 - 8$$

$$\int f'(x)dx = \int (6x^2 - 8)dx$$

$$f(x) = 2x^3 - 8x + C \text{—————(1)}$$

$$xf(x) = \frac{3}{2}x^4 - 4x^2 + 6 + \int_2^x f(t)dt$$

$$2f(2) = 24 - 16 + 6 + 0$$

$$f(2) = 7 \text{ 代入 (1), } 7 = 16 - 16 + C$$

$$C = 7$$

$$\therefore f(x) = 2x^3 - 8x + 7$$